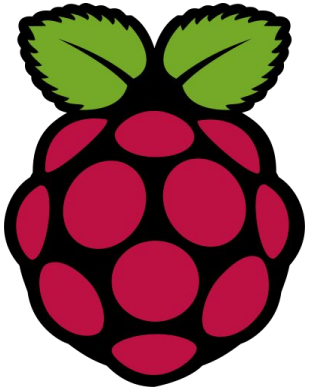
Mastering Raspberry Pi: Hands-On Practical Applications for Beginners



SINGAPORE WORKFORCE SKILLS QUALIFICATIONS

Tertiary Infotech Academy

Course Development | Training | Agentic AI Automation

Version: 14.0

Trainer: _____

Course Code: TGS-2020505433

Website: www.tertiarycourses.com.sg

Email: enquiry@tertiaryinfotech.com

Tel: +65 6100 0613

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Digital Attendance

- It is **mandatory** for you to take both AM, PM and Assessment digital attendance for WSQ funded courses.

- The trainer or administrator will show you the digital attendance QR code generated from SSG portal.

- Please scan the QR code from **your mobile phone camera** and submit your attendance.

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About the Trainer

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Let's Know Each Other...

Ground Rules

- Set your mobile phone to silent mode
- Participate actively in the class. No question is stupid.
- Mutual respect. Agree to disagree.
- One conversation at one time.
- Be punctual. Back from breaks on time.
- Exit the class silently if you need to step out for phone call, toilet break etc.
- 75% attendance is required

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Lesson Plan

Day 1

- **Digital Attendance (AM)**
- Trainer and Learners Introduction
- Learning Outcomes
- Course Outline
- Topic 1: Overview of Internet of Things (IoT)
- Lunch Break
- **Digital Attendance (PM)**
- Topic 2: Collect and Post Environmental Data to Cloud
- End of Day 1

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Lesson Plan

Day 2

- **Digital Attendance (AM)**
- Topic 3: IoT Data Analytics and Visualization
- Lunch Break
- **Digital Attendance (PM)**
- Topic 4: Monitor Data to Trigger Control from Cloud
- **Course Feedback and TRAQOM Survey**
- **Digital Attendance (Assessment)**
- Final Assessment
- End of Class

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Skills Framework

TSC Title: Internet of Things Management
TSC Code: EVS-TEM-4002-1.1 TSC

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Skills Framework TSC

TSC Abilities:
A1: Analyse relevant data to recommend control actions
A2: Identify areas for implementing IoT solutions for process improvement
A3: Use relevant tool to analyse and record performance
A4: Monitor the effectiveness of IoT solutions

TSC Knowledge:
K1: IoT concept and technical knowledge of IoT implementation
K2: Connectivity in devices using sensors, smart devices and other technologies for data collection and communication
K3: Equipment automation
K4: Building and/or Environment automation
K5: Advanced process control
K6: Security and privacy applications for IoT
K7: IoT guidelines and communication standards

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Learning Outcomes

- LO1: Analyze the uses and functions of IoT technologies
- LO2: Collect environment data and post the data to cloud computing
- LO3: Use cloud computing to analyse and visualize IoT data
- LO4: Monitor data and trigger control on devices

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Course Outline

Topic 1: Overview of Internet of Things (IoT) (A1, K1)

- **What is IoT?**
- **Sensors and Actuators for IoT**
- **Wireless Communication Technologies for IoT**
- **IoT Applications and Use Cases**

Topic 2: Collect and Post Environmental Data to Cloud (A2, K2, K3)

- **What is Cloud Computing**
- **Setup Cloud Computing Account**
- **Collect Environmental Data with Sensors**
- **Transmit Environmental Data using Raspberry Pi Pico W**
- **Post Data to Cloud using MQTT API**

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Course Outline

Topic 3: IoT Data Analytics and Visualization (A3, K4, K5)

- **Analyze IoT Data on Cloud**
- **Visualize IoT Data on Cloud**

Topic 4: Monitor Data to Trigger Control from Cloud (A4, K6, K7)

- **Read Data using MQTT API**
- **Trigger Control on Devices**
- **IoT Security**

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Final Assessment

- **Written Assessment (Short Answer Questions)**
- **Practical Performance**

- Assessment format: Open Book
- Open book assessment ONLY includes Slides, Learner Guide or any approved materials.
- Appeal process

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Briefing for Assessment

- Place phones & other materials under the table or on the floor
- No photos or recording of assessment scripts
- No discussion during assessment
- Use black/blue pen for assessment [hard copies]
- No usage of liquid paper or correction tape
- Assessment scripts will be collected when the time is up

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Criteria for Funding

- Minimum attendance rate of 75% based on SSG Digital Attendance record.
- Complete the assessment and be assessed as ‘Competent’

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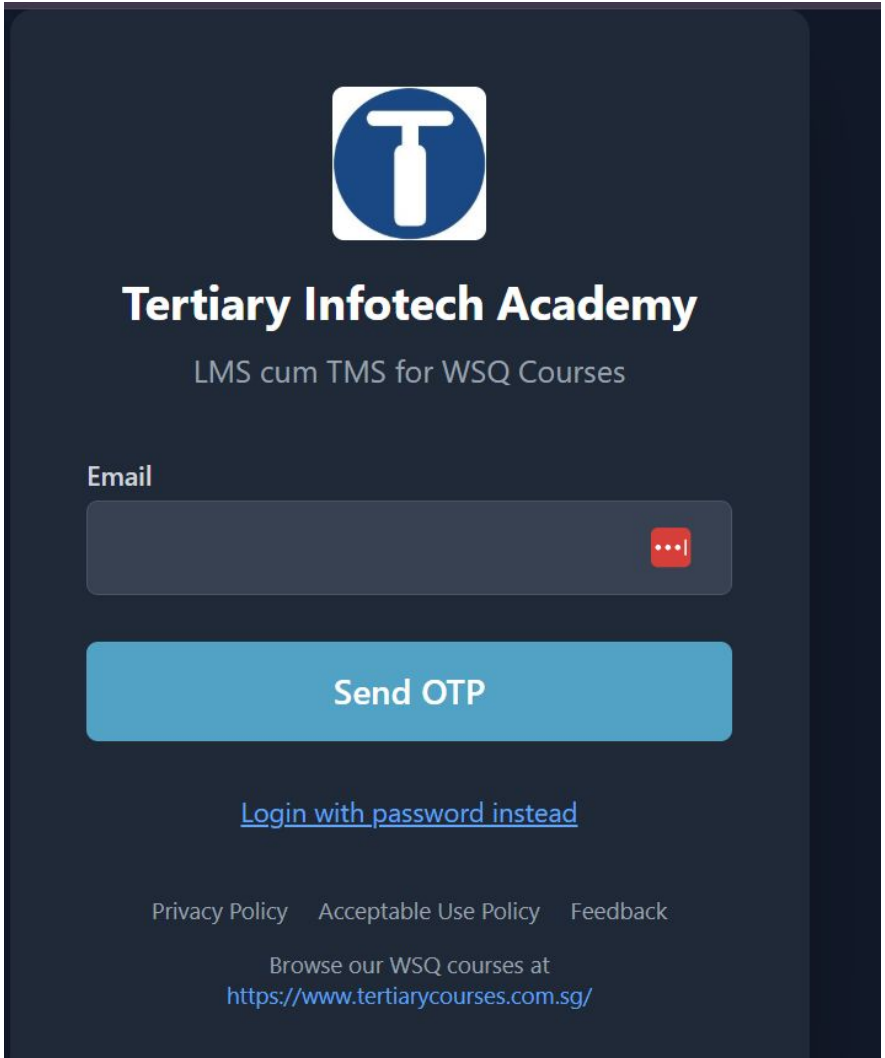
LMS/TMS

https://lms-tms.tertiaryinfotech.com



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Topic 1
Overview of Internet of Things (IoT)

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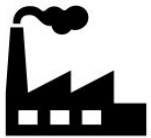
4th Industrial Revolution

Manufacturing continues to evolve and has progressed through four stages:

1.0: Steam power to mechanization
2.0: Mass production and powered by electricity
3.0: Computers and automation
4.0: Cyber-physical systems to enable the computerization.



Industry 1.0
Mechanization, steam power, weaving loom

(18th century, 1760 - 1840)

Industry 2.0
Mass production, assembly line, electrical energy

(19th century, 1871 - 1914)

Industry 3.0
Automation, computers and electronics

(late 20th century, 1970 - present)

Industry 4.0
Cyber Physical Systems, internet of things, networks

(early 21st century, 2011 - present)

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4th Industrial Revolution

- Emphasizes heavily on automation, machine learning, real-time data, and interconnectivity
- Smart manufacturing
- Synchronizes physical production and operations with Smart Digital Technology

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4th Industrial Revolution

- Access and stay connected to real-time insights across products, people, partners, and processes.
- Advanced tools and techniques for improving the efficiency of manufacturing operations
- Revolutionizing the way every single operation of organization.

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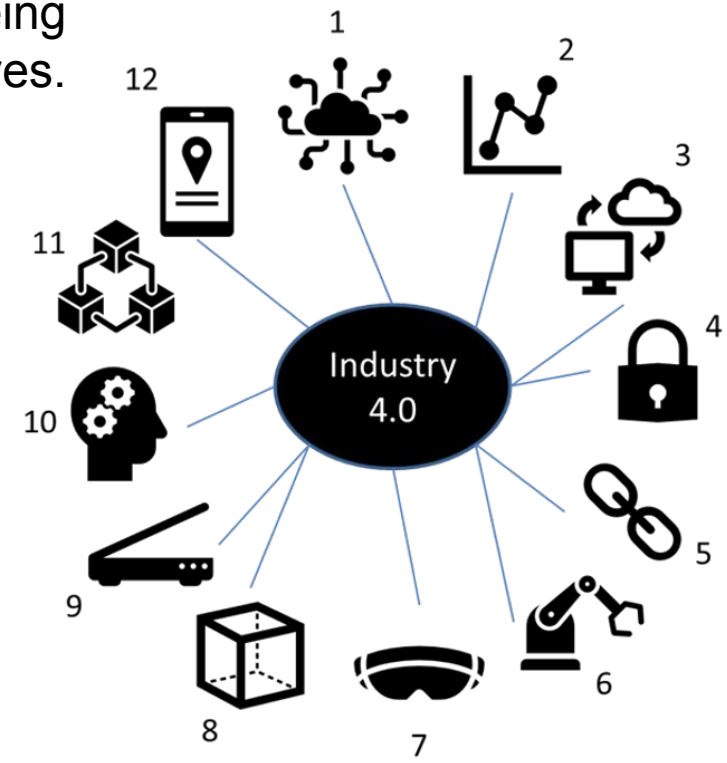
4th Industrial Revolution

There are many technologies and applications being used in Industry 4.0 to achieve goals and objectives.

- Industrial Internet of Things (IIoT)
- Big Data and Analytics
- Cloud Computing
- Cybersecurity
- Horizontal & Vertical Integration
- Autonomous Robots
- Augmented Reality
- Additive Manufacturing
- Simulation or Digital Twin
- Artificial Intelligence
- Blockchain Technology
- 5G Technology



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Internet of Things (IoT)

IoT (Internet of Things) refers to the network of physical devices, vehicles, buildings, and other objects that are embedded with sensors, software, and connectivity, allowing them to collect and exchange data with each other and with central servers or the cloud.

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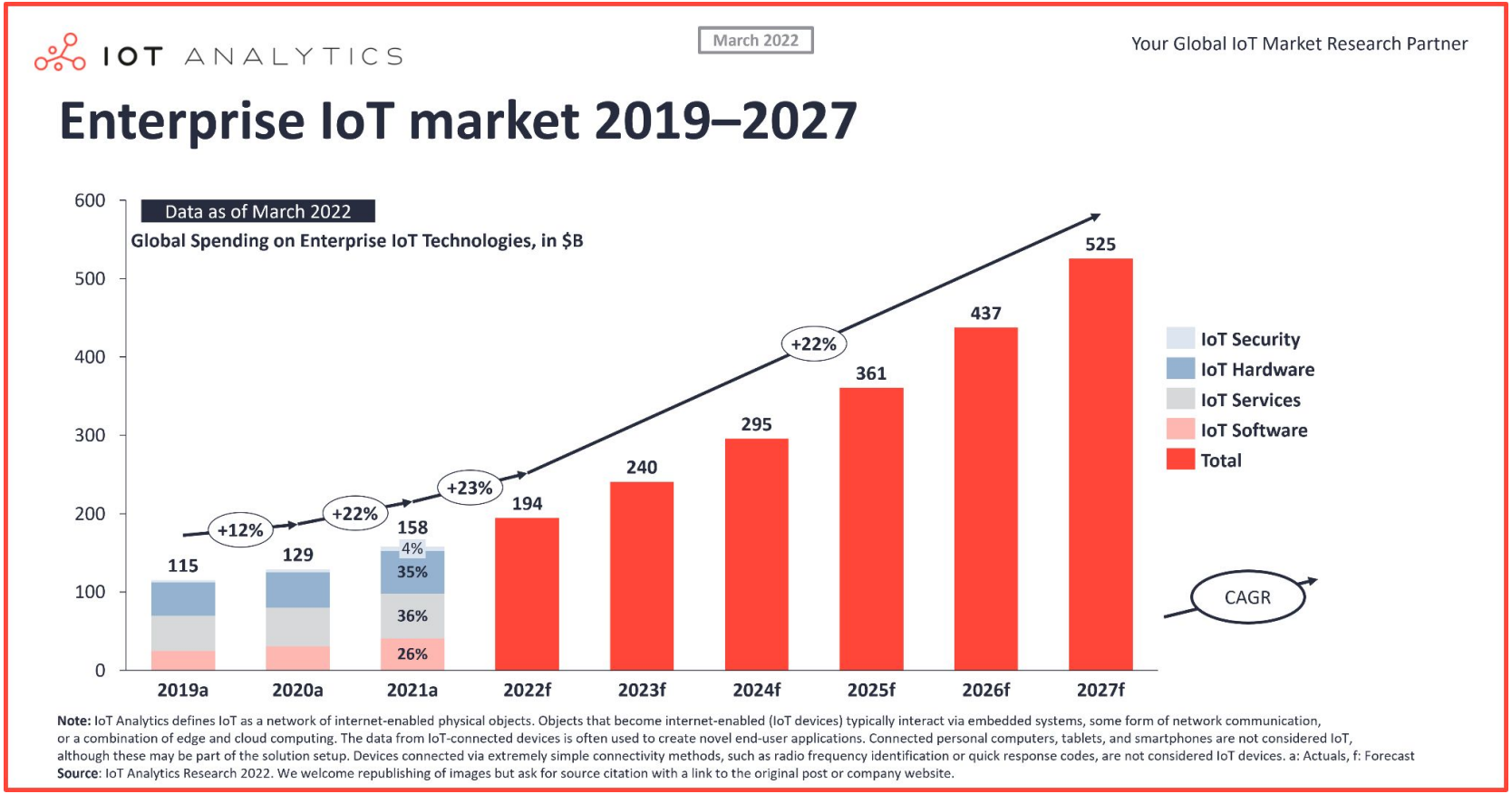
Internet of Things (IoT)



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Driving Forces of IoT

- Low cost sensors and long range wireless technology are instrumental to IoT
- 5G will drive IoT to mass adoption.

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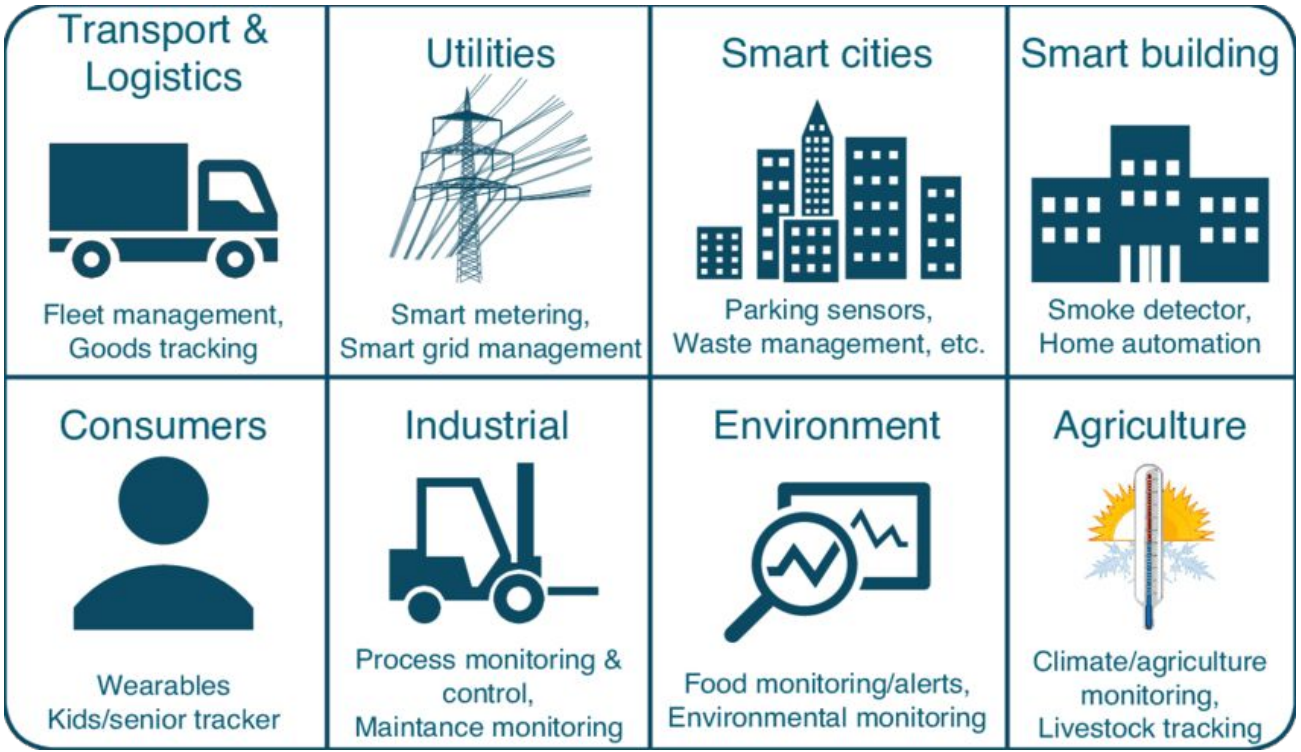
IoT Applications

IoT based application has found its way into every conceivable field, from agriculture to aerospace



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IoT Use Case - Healthcare

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IoT Use Case - Healthcare

- Doctors, nurses, and orderlies often need to know the exact location of patient-assistance assets such as wheelchairs.
- When a hospital's wheelchairs are equipped with IoT sensors, they can be tracked from the IoT asset-monitoring application so that anyone looking for one can quickly find the nearest available wheelchair.
- Many hospital assets can be tracked this way to ensure proper usage as well as financial accounting for the physical assets in each department.

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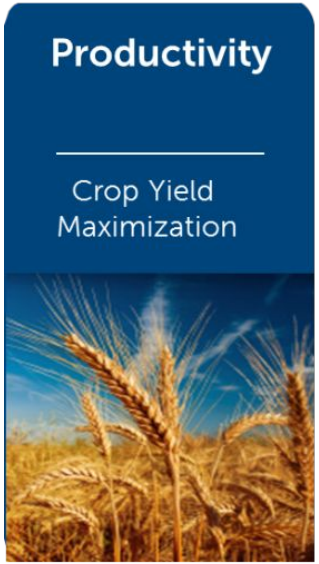
IoT Use Case - Agriculture



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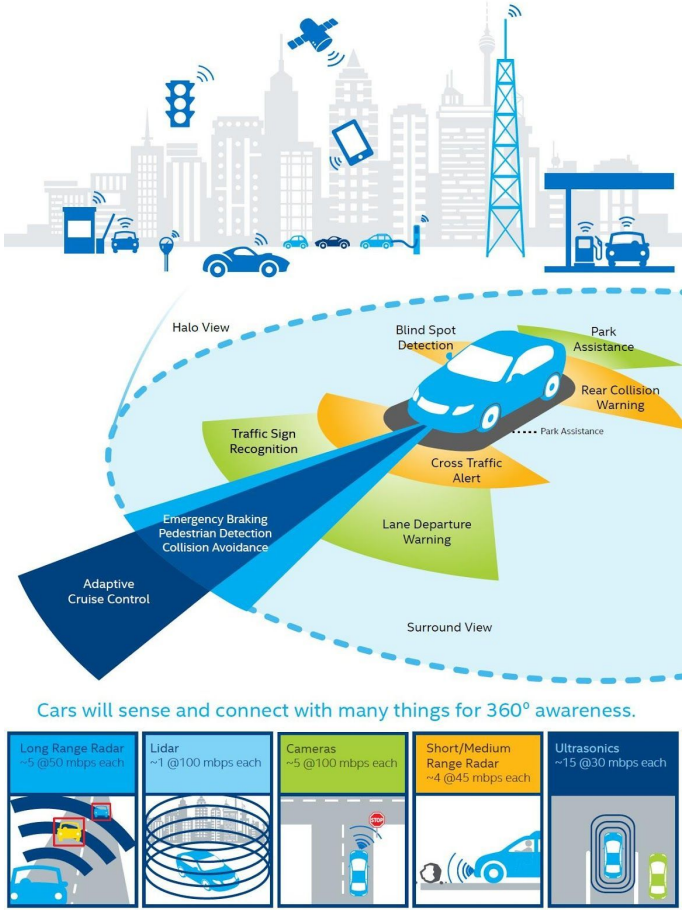
IoT Use Case - Connected Vehicles



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IoT Use Case - Connected Vehicles

- There are many ways vehicles, such as cars, can be connected to the internet. It can be through smart dashcams, infotainment systems, or even the vehicle's connected gateway. They collect data from the accelerator, brakes, speedometer, odometer, wheels, and fuel tanks to monitor both driver performance and vehicle health.
- Connected cars have a range of uses:
  - Monitoring rental car fleets to increase fuel efficiency and reduce costs.
  - Helping parents track the driving behavior of their children.
  - Notifying friends and family automatically in case of a car crash.
  - Predicting and preventing vehicle maintenance needs.

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IoT Use Case - Smart Home

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IoT Use Case - Smart Home

- Smart home devices are mainly focused on improving the efficiency and safety of the house, as well as improving home networking. Devices like smart outlets monitor electricity usage and smart thermostats provide better temperature control.
- Home security systems like door locks, security cameras, and water leak detectors can detect and prevent threats, and send alerts to homeowners.
- Connected devices for the home can be used for:
  - Automatically turning off devices not being used.
  - Rental property management and maintenance.
  - Finding misplaced items like keys or wallets.
  - Automating daily tasks like vacuuming, making coffee, etc.

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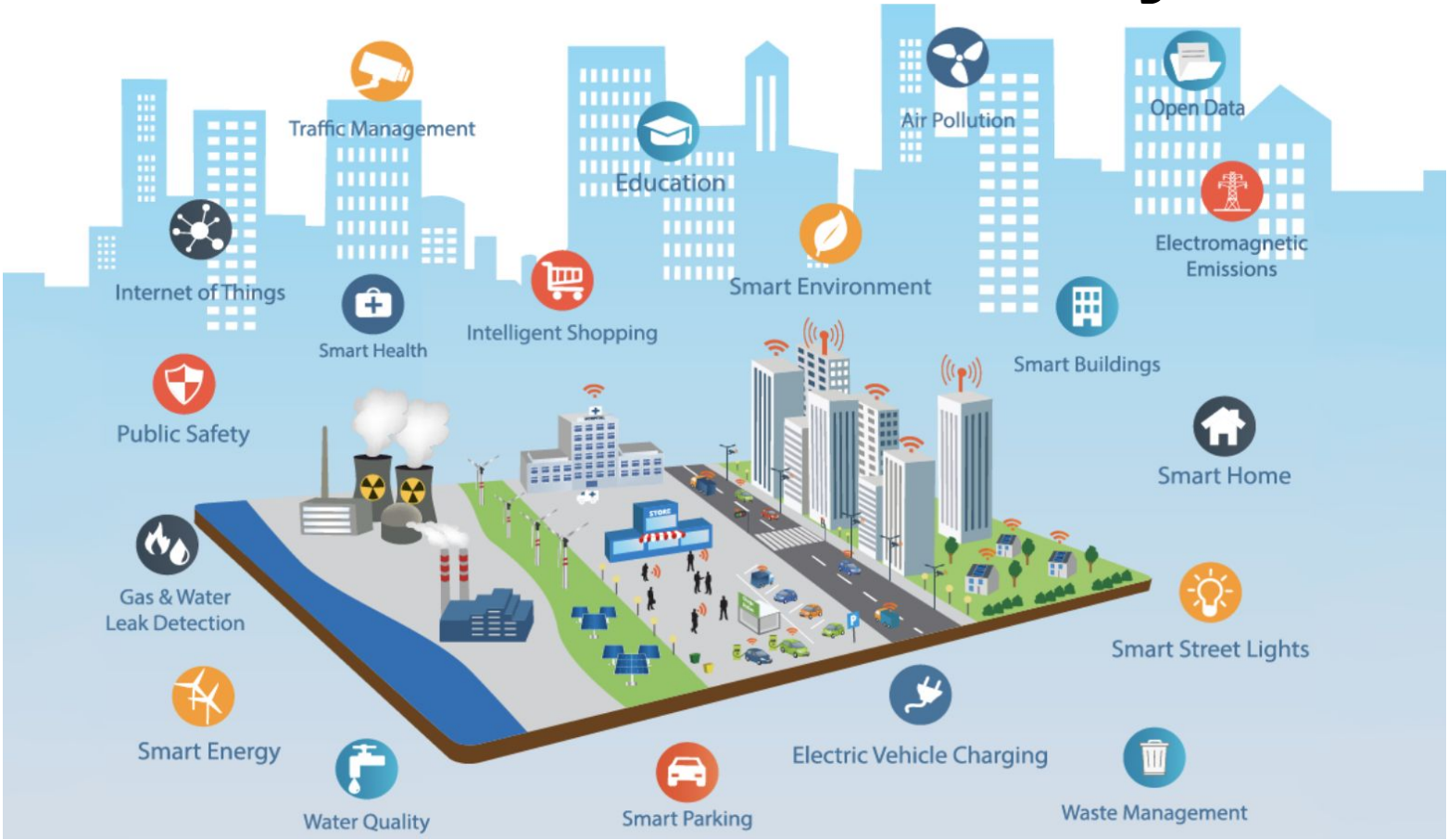
Use Case - Smart City



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IoT Use Case - Smart Home

- IoT applications have made urban planning and infrastructure maintenance more efficient. Governments are using IoT applications to tackle problems in infrastructure, health, and the environment.
- IoT applications can be used for:
  - Measuring air quality and radiation levels.
  - Reducing energy bills with smart lighting systems.
  - Detecting maintenance needs for critical infrastructures such as streets, bridges, and pipelines.
  - Increasing profits through efficient parking management.

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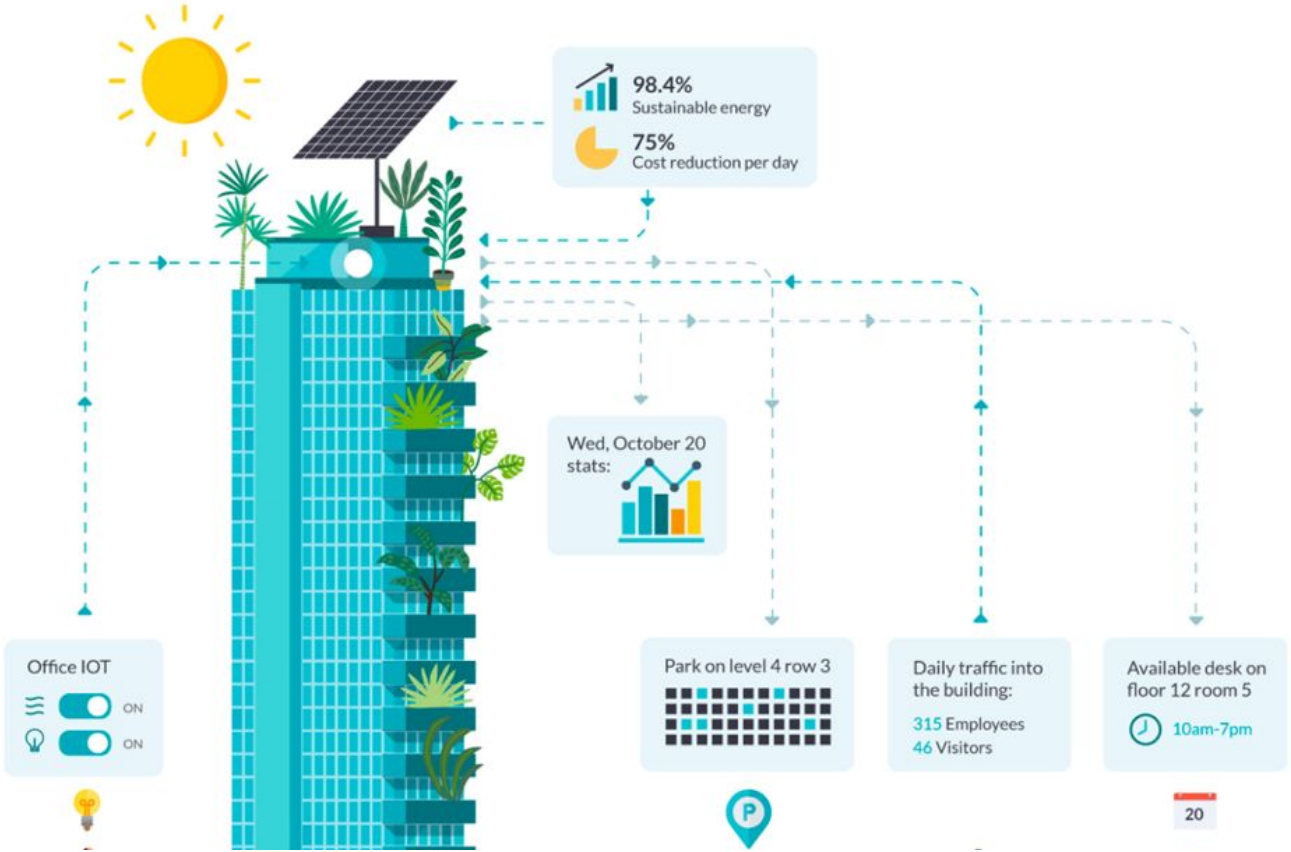
IoT Use Case - Smart Building



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IoT Use Case - Smart Building

- Buildings such as college campuses and commercial buildings use IoT applications to drive greater operational efficiencies.
- IoT devices can be use in smart buildings for:
  - Reducing energy consumption.
  - Lowering maintenance costs.
  - Utilizing work-spaces more efficiently.

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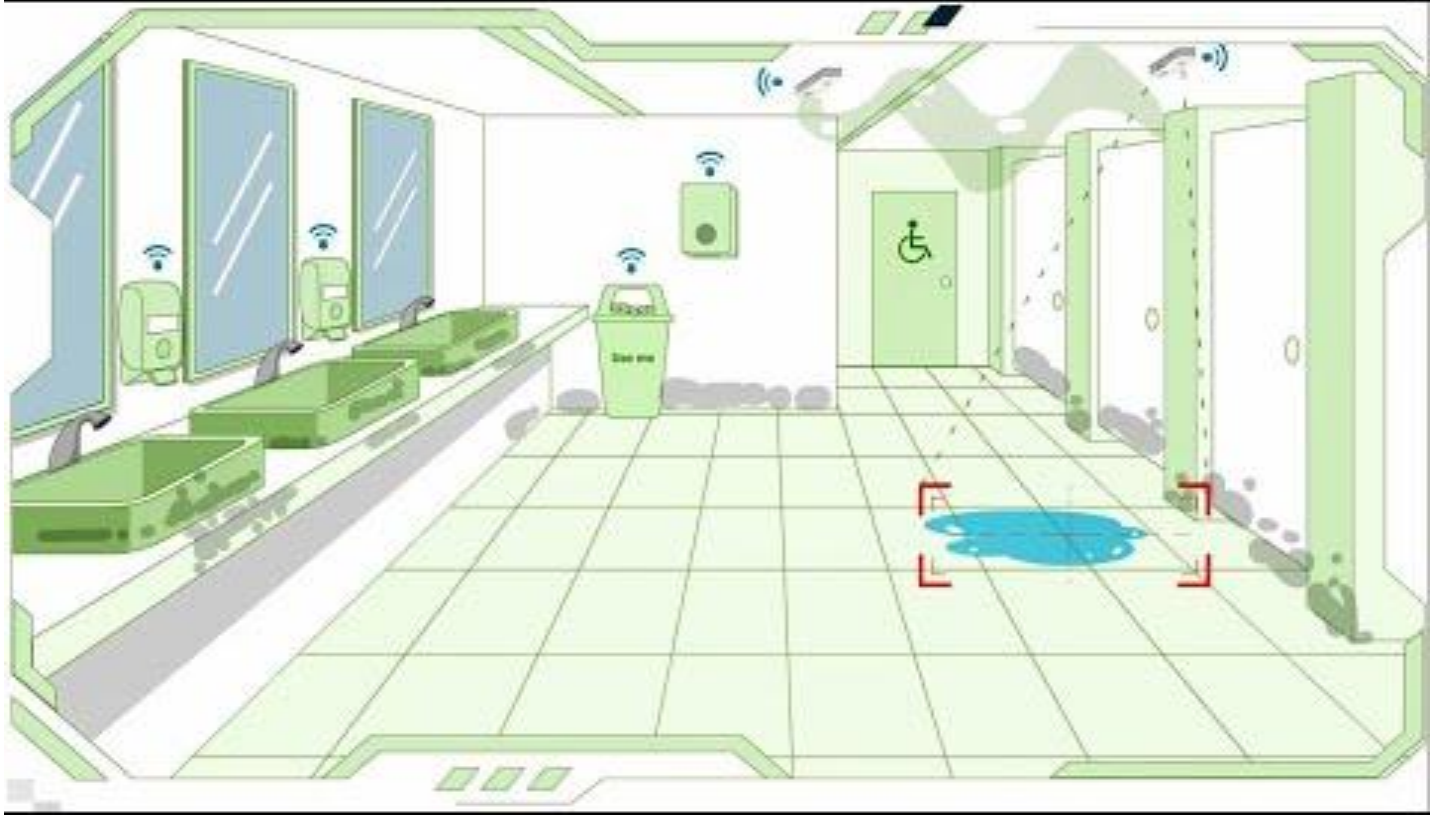
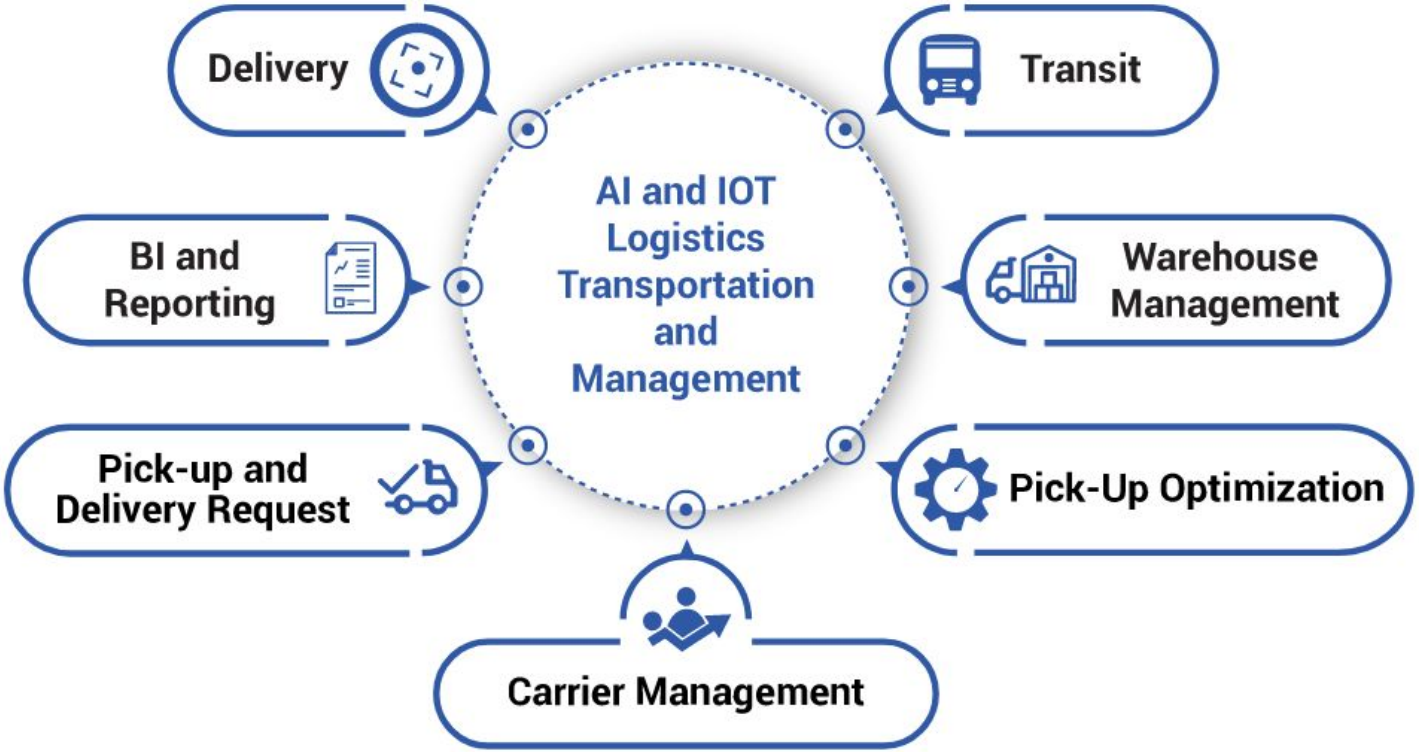
IoT Use Case - Smart Toilet



IoT Use Case - Logistics



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IoT Use Case - Logistics

- Fleets of cars, trucks, ships, and trains that carry inventory can be rerouted based on weather conditions, vehicle availability, or driver availability, thanks to IoT sensor data.
- The inventory itself could also be equipped with sensors for track-and-trace and temperature-control monitoring.
- The food and beverage, flower, and pharmaceutical industries often carry temperature-sensitive inventory that would benefit greatly from IoT monitoring applications that send alerts when temperatures rise or fall to a level that threatens the product.

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IoT Use Case - Manufacturing

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IoT Use Case - Manufacturing

- Manufacturers can gain a competitive advantage by using production-line monitoring to enable proactive maintenance on equipment when sensors detect an impending failure.
- Sensors can measure when production output is compromised. With the help of sensor alerts, manufacturers can quickly check equipment for accuracy or remove it from production until it is repaired.
- This allows companies to reduce operating costs, get better uptime, and improve asset performance management.

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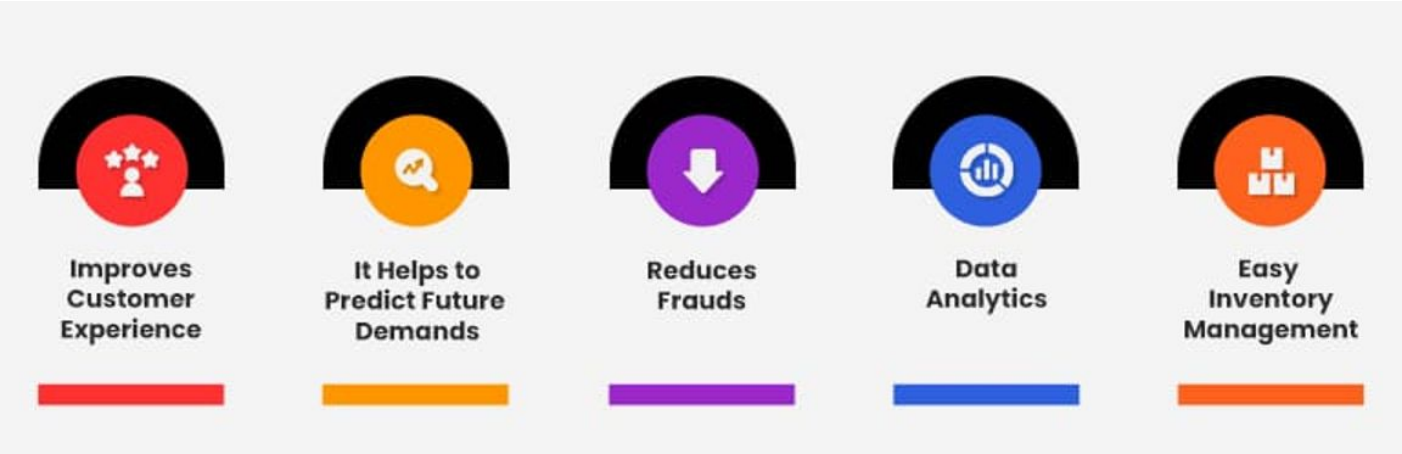
IoT Use Case - Retail

- IoT applications allow retail companies to manage inventory, improve customer experience, optimize supply chain, and reduce operational costs.
- Smart shelves fitted with weight sensors can collect RFID-based information and send the data to the IoT platform to automatically monitor inventory and trigger alerts if items are running low.
- Beacons can push targeted offers and promotions to customers to provide an engaging experience.



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Types of IoT

- Consumer IoT - Primarily for everyday use. Eg: home appliances, voice assistance, and light fixtures.
- Commercial IoT - Primarily used in the healthcare and transport industries. Eg: smart pacemakers and monitoring systems, connected cars
- Industrial Internet of Things (IIoT) - Primarily used with industrial applications, such as in the manufacturing and energy sectors. Eg: Digital control systems, smart agriculture and industrial big data.
- Infrastructure IoT - Primarily used for connectivity in smart cities. Eg: infrastructure sensors and management systems.

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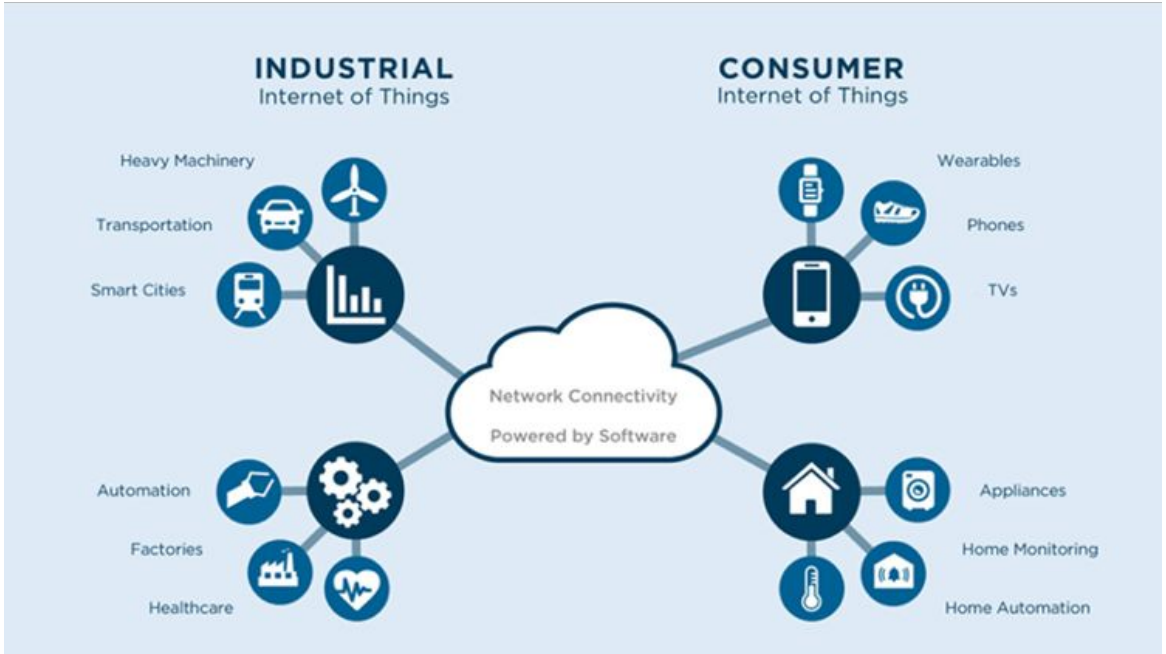
What is Industrial IoT?

The industrial internet of things (IIoT) refers to the extension and use of the internet of things (IoT) in industrial sectors and applications.



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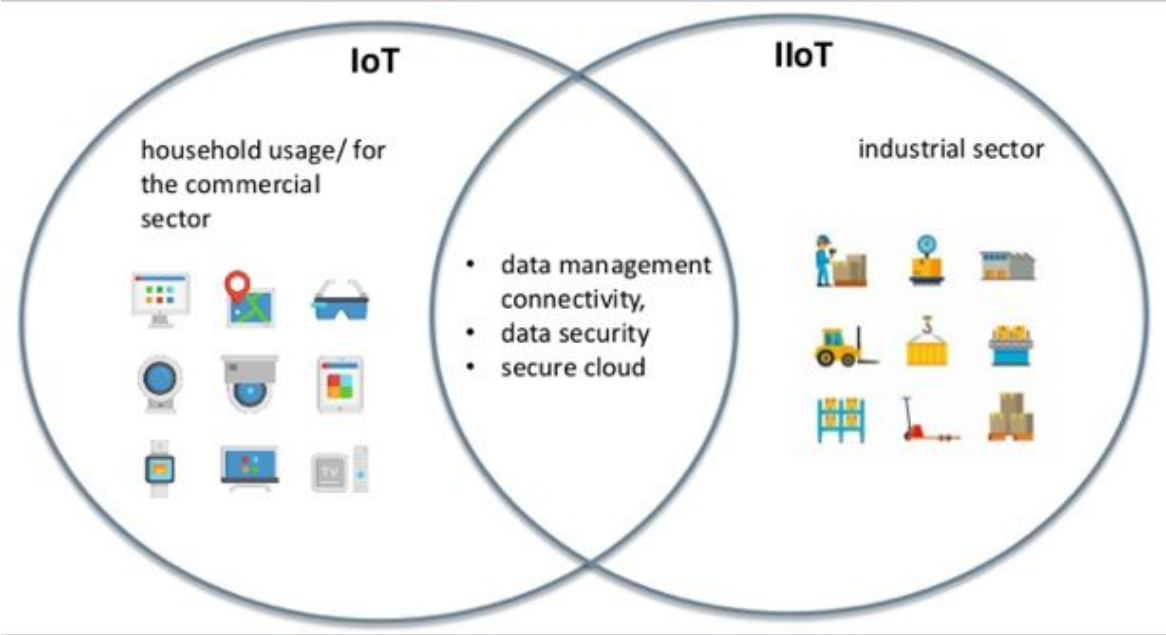
IoT vs IIoT

Industrial devices, from sensors to equipment, give business owners detailed, real-time data that can be used to improve business processes.



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IoT Implementation

- **Hardware:** IoT devices typically require hardware components such as sensors, processors, and communication modules that can connect to the internet or other networks.
- **Software:** IoT devices also require software that can interpret and process the data collected by the sensors and communicate with other devices or servers.
- **Communication protocols:** IoT devices use different communication protocols, such as Wi-Fi, Bluetooth, Zigbee, and cellular networks, to communicate with each other and with central servers or the cloud.
- **Data analytics:** The data collected by IoT devices needs to be processed and analyzed to derive meaningful insights that can be used for decision-making.

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IoT Implementation

- **Security:** IoT devices and networks need to be secured against cyber attacks, data breaches, and other security threats.
- **Cloud computing:** IoT data is often stored and processed in the cloud, which requires knowledge of cloud computing platforms, such as AWS, Azure, or Google Cloud.
- **Artificial Intelligence and Machine Learning:** IoT systems can use AI and ML to improve decision-making, predictive maintenance, and automate tasks.
- **Energy efficiency:** IoT devices often operate on battery power or other low-power sources, which requires knowledge of energy-efficient design and management.

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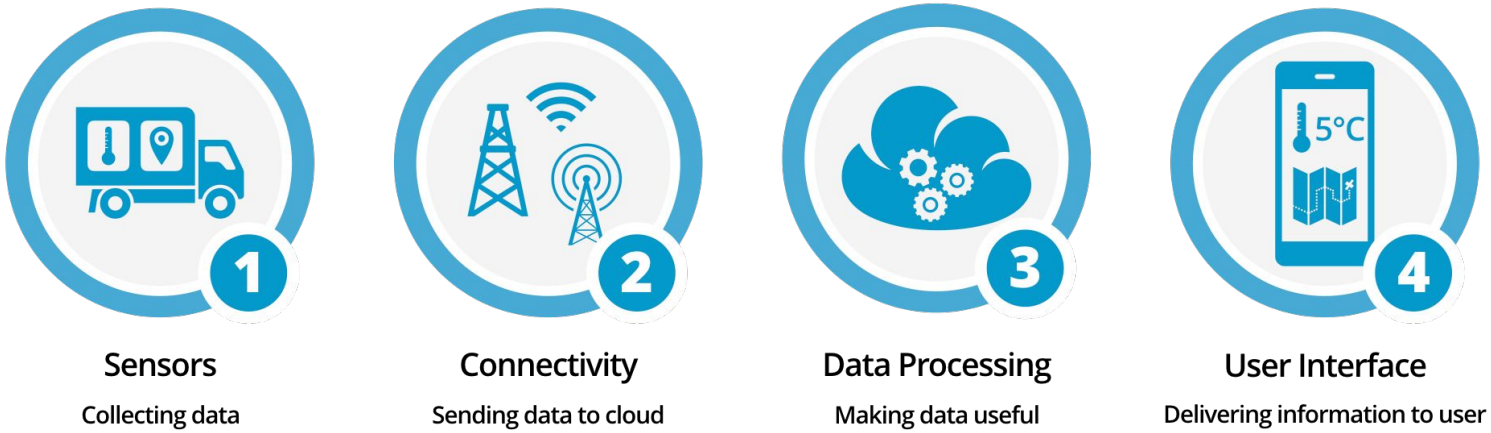
IoT Components



A “thing” is any object with embedded electronics that can transfer data over a network — without any human interaction.

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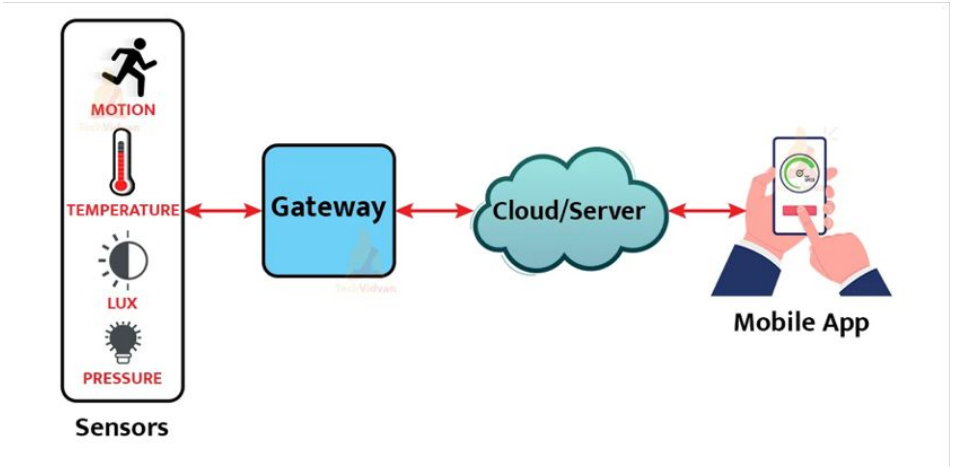
How does IoT work?

- Devices and objects with built in sensors are connected to an Internet of Things platform, which integrates data from the different devices and applies analytics to share the most valuable information with applications built to address specific needs.
- These powerful IoT platforms can pinpoint exactly what information is useful and what can safely be ignored. This information can be used to detect patterns, make recommendations, and detect possible problems before they occur.



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What Are IoT Devices?

IoT devices are hardware devices, such as sensors, gadgets, appliances and other machines that collect and exchange data over the Internet.

They are programmed for certain applications and can be embedded into other IoT devices.

For example, an IoT device in your car can identify the traffic ahead and send out a message automatically to the person you are about to meet of your impending delay.

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Functionalities of IoT devices

- Different IoT devices have different functions, but they all have similarities in terms of how they work.
- Firstly, IoT devices are physical objects that sense things going on in the physical world. They contain an integrated CPU, network adapter and firmware, and are usually connected to a Dynamic Host Configuration Protocol server. It also requires an IP address to function over the network.

- Most IoT devices are configured and managed through a software application. For example, an app on your smartphone to control the lights in your home.
- Some devices also have integrated web servers, which eliminates the need for external applications. For example, the lights switch on immediately when you enter a room.

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Examples of IoT Devices

- Home Security - The key driver behind smart and secure homes is IoT. A variety of sensors, lights, alarms and cameras (all of which can be controlled from a smartphone) are connected via IoT to provide 24x7 security.
- Activity Trackers - Smart home security cameras provide alerts and peace of mind. Activity trackers are sensor devices that can monitor and transmit key health indicators in real-time. You can track and manage your blood pressure, appetite, physical movement and oxygen levels.
- Industrial Security and Safety - IoT-enabled detection systems, sensors and cameras can be placed in restricted areas to detect trespassers. They can also identify pressure buildups and small leaks of hazardous chemicals and fix them before they become serious problems.
- Motion Detection - Motion sensors can detect vibrations in buildings, bridges, dams and other large-scale structures. These devices can identify anomalies and disturbances in the structures that could lead to catastrophic failures. They can also be used in areas susceptible to floods, landslides, and earthquakes.

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What is Sensor?

- A sensor is a device that produces an output signal for the purpose of sensing a physical phenomenon. In the broadest definition, a sensor is a device, module, machine, or subsystem that detects events or changes in its environment and sends the information to other electronics, frequently a computer processor.
- There are different types of sensors, for example, consider a thermocouple which can be considered as a temperature sensor that produces an output voltage based on the input temperature changes.

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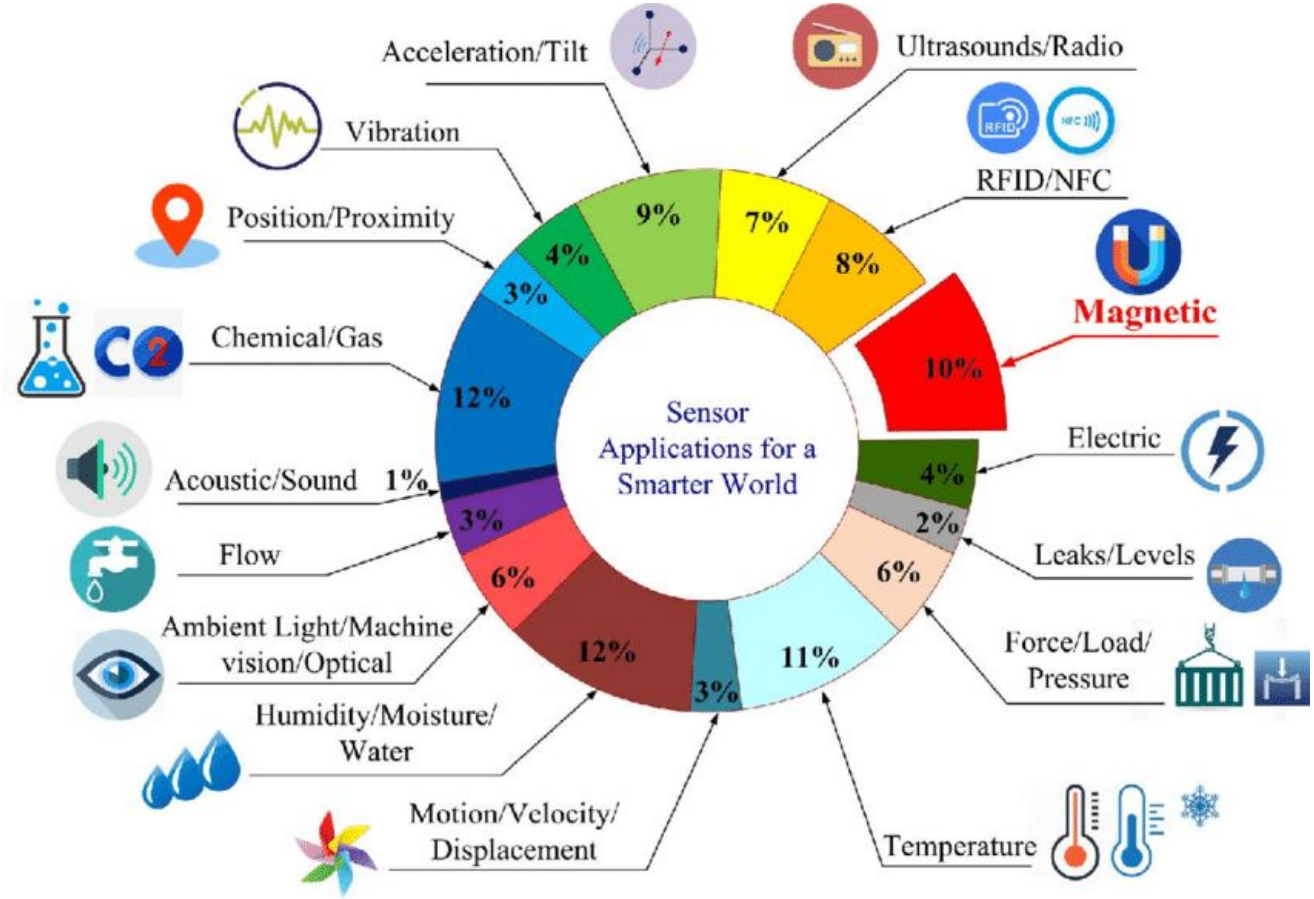
Sensors for IoT



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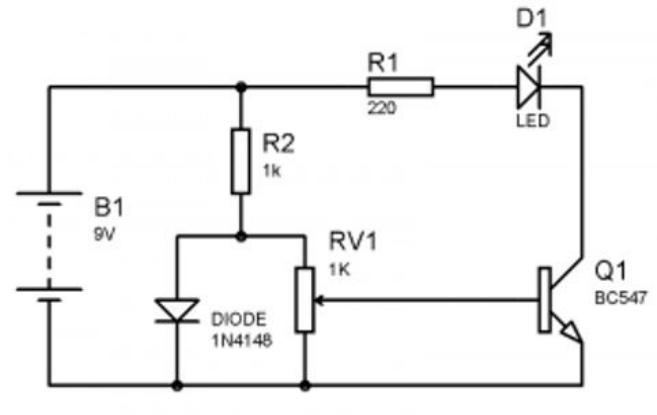
Temperature Sensor

- Temperature is one of the most commonly measured environmental quantities for different reasons.
- There are different types of temperature sensors that can measure temperature, such as a thermocouple, thermistors, semiconductor temperature sensors, resistance temperature detectors (RTDs), and so on.



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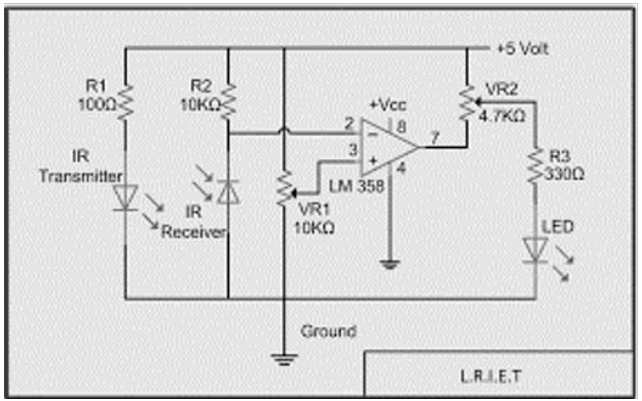
IR Sensor

- The small photo chips having a photocell which are used to emit and detect the infrared light are called as IR sensors.
- IR sensors are generally used for designing remote control technology. IR sensors can be used for detecting obstacles of the robotic vehicle and thus control the direction of the robotic vehicle.



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Proximity Sensors

- This a type of IoT sensor where it identifies the existence or non-existence of the surrounding object or finds the object properties.
- Then it converts the detected signal into the form that is clearly understood by the user or might be a simple electronic device that gets in no contact with them.

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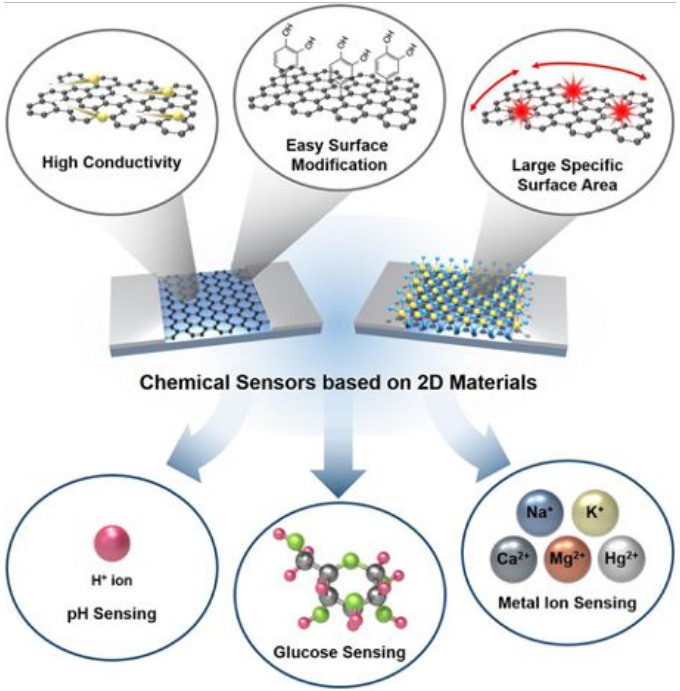
Chemical Sensors

- The main objective of these sensors is to signify any kind of changes in the liquid or to detect any air chemical variations.
- These are crucially implemented in bigger towns and cities because it is important to look for changes and provide safety for the population.



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Gas Sensors

- These are almost the same as chemical sensors but are exclusively implemented to observe modifications of the quality of the air and to find out the existence of different types of gases.

- Examples:
  - Hydrogen type
  - Ozone monitoring type
  - Hygrometer
  - Carbon-dioxide sensor
  - Electrochemical gaseous type
  - Catalytic bead type

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Humidity Sensors

Humidity is the term that is specified as the amount of vapor that exists in the atmospheric air or in other gaseous substances.

Humidity sensors generally adhere to the utilization of temperature sensors as because most of the manufacturing operations need exact operating conditions.

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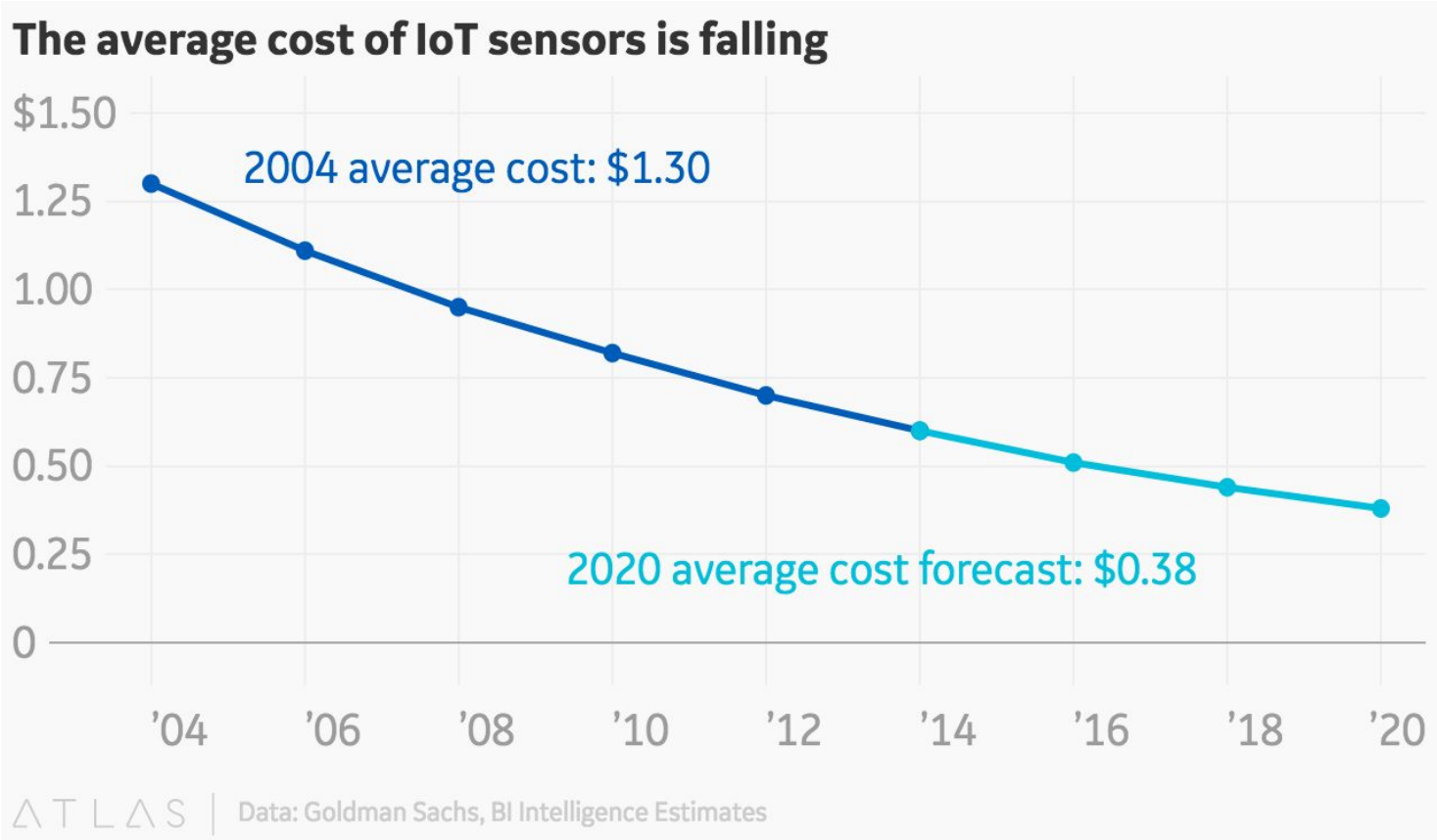
Sensors for IoT



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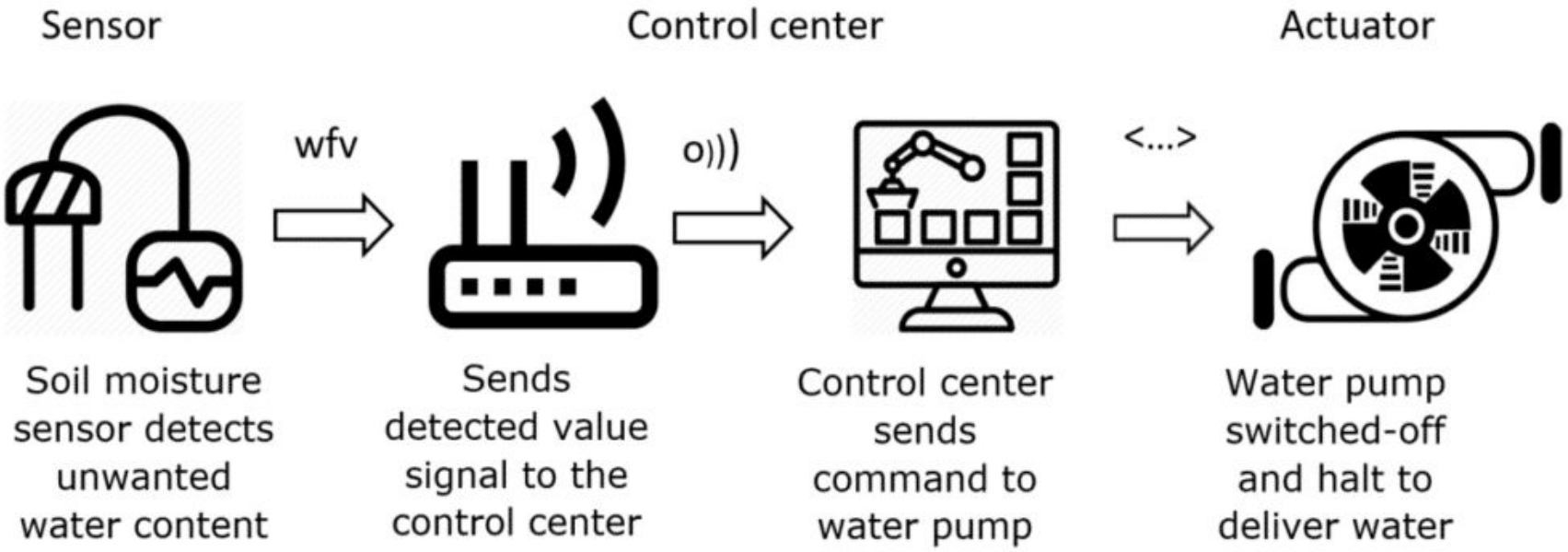
IoT Actuator



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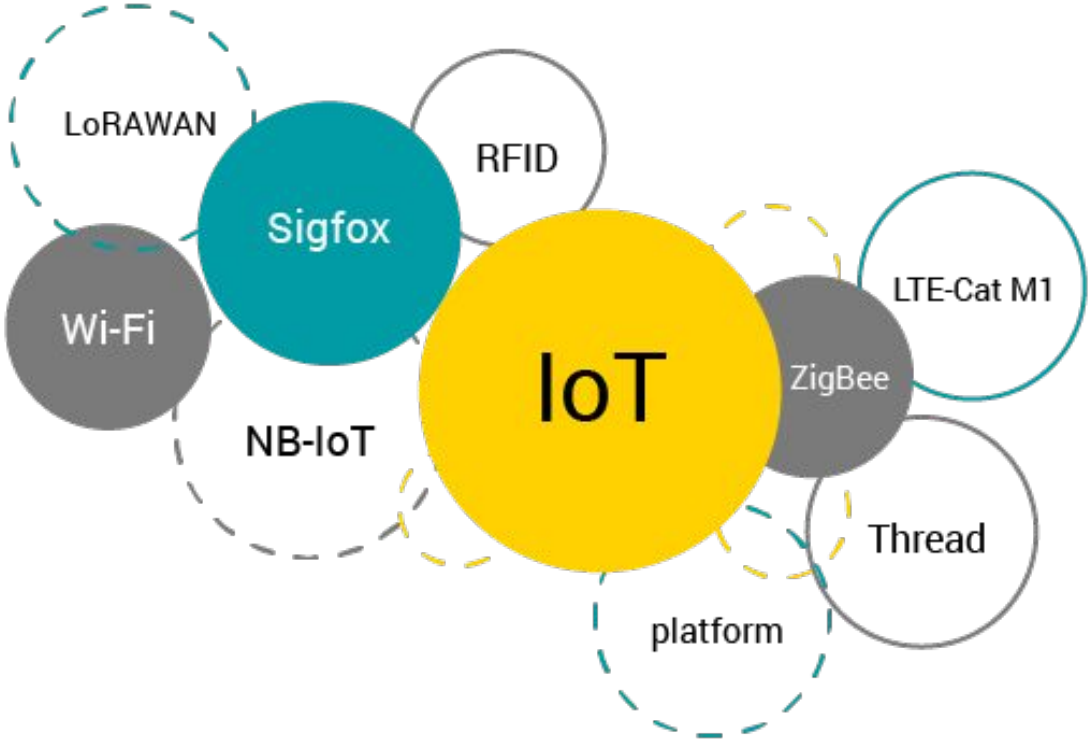
Wireless Technologies for IoT



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Wireless Technologies for IoT

- Wi-Fi: Wi-Fi is a widely-used wireless communication technology that offers high data rates and reliable connectivity over short to medium distances.
- Bluetooth: Bluetooth is a low-power, short-range wireless technology that is commonly used for connecting IoT devices to smartphones, tablets, and other devices.
- Zigbee: Zigbee is a low-power, wireless mesh networking technology that is designed for IoT applications that require low data rates, low power consumption, and reliable communication over longer distances.
- Z-Wave: Z-Wave is a wireless communication technology that is optimized for low-power IoT devices and offers reliable communication over longer distances.

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Wireless Technologies for IoT

- LoRaWAN: LoRaWAN is a long-range, low-power wireless communication technology that is designed for IoT applications that require long-range communication, such as smart city applications.
- Sigfox: Sigfox is a low-power, wide-area network (LPWAN) technology that is designed for IoT applications that require low data rates and long-range communication, such as asset tracking and remote monitoring.
- NB-IoT: Narrowband IoT (NB-IoT) is a low-power, wide-area network (LPWAN) technology that is designed for IoT applications that require low data rates, long-range communication, and reliable connectivity.

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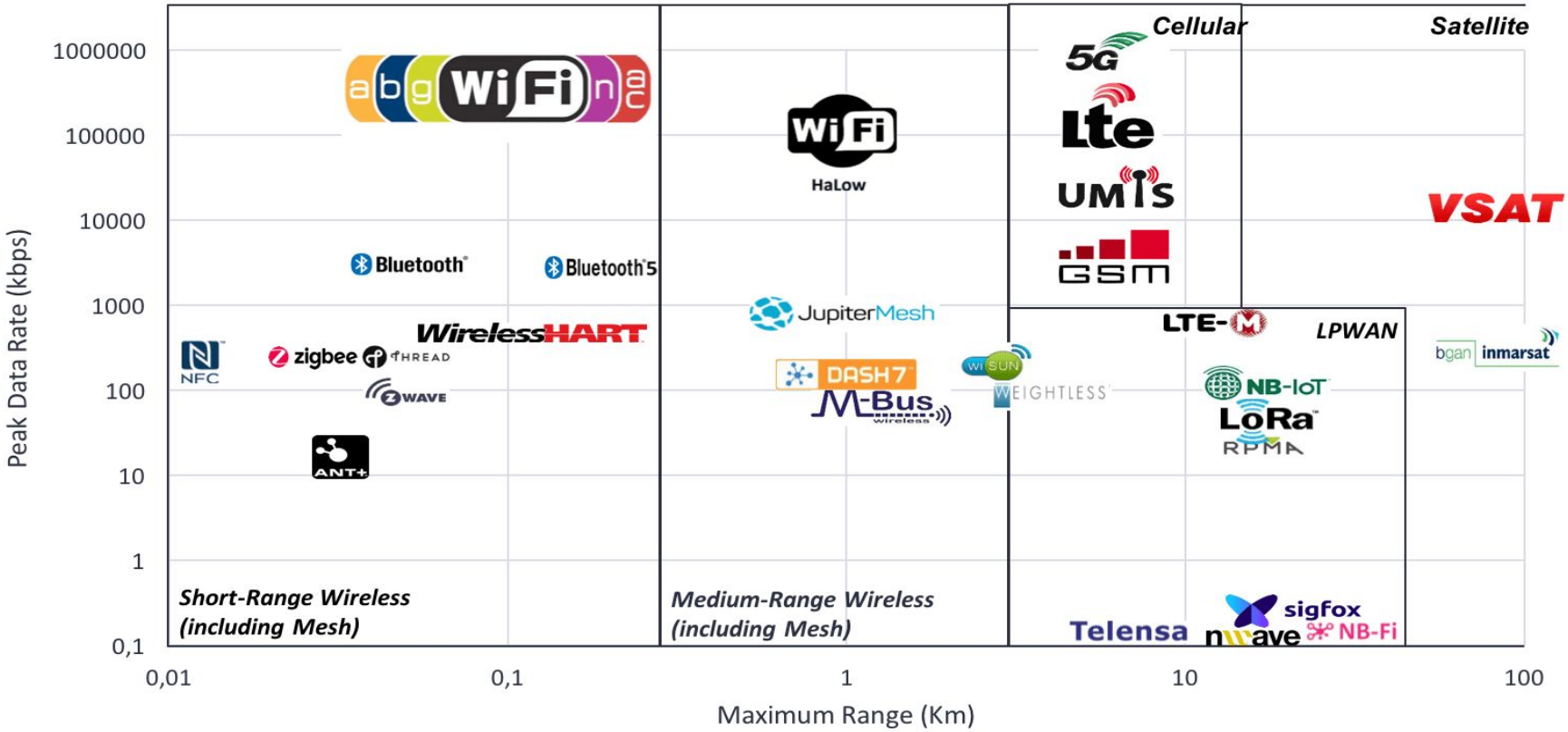
Comparison of Wireless Technologies



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Topic 2
Collect and Post Data to Cloud

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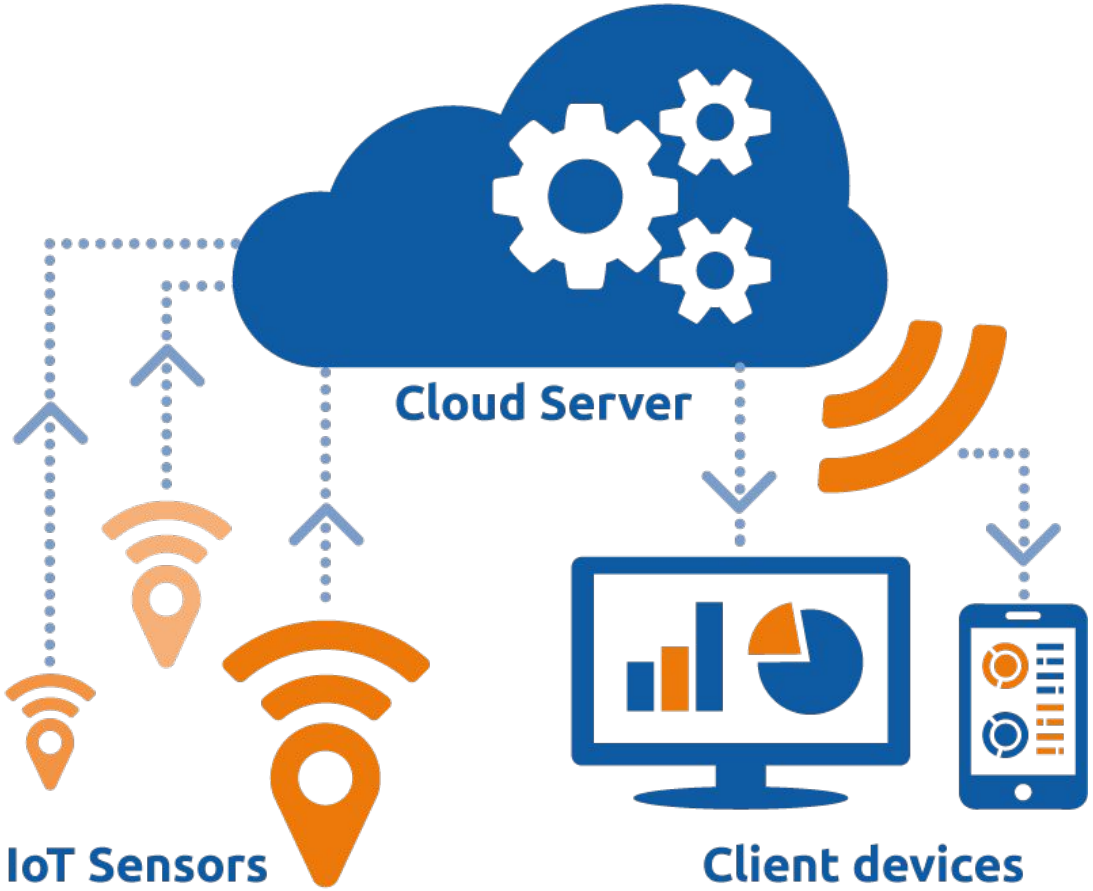
Cloud Computing



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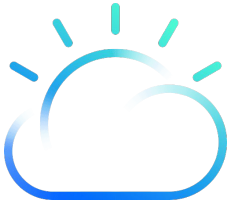
Cloud Platforms



Google Cloud

Alibaba Cloud

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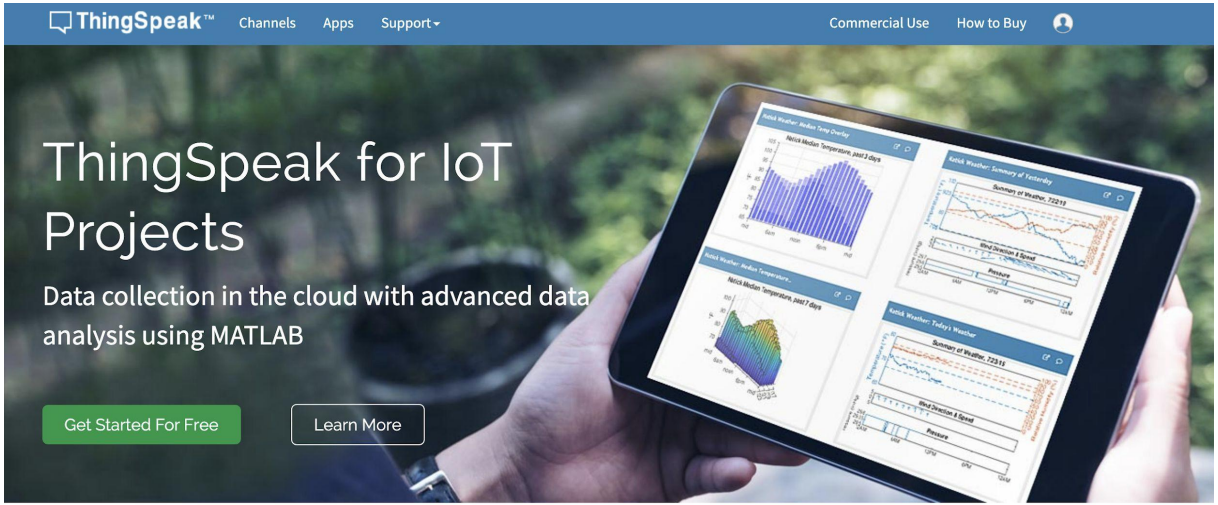
ThingSpeak

- ThingSpeak™ is an IoT analytics platform service that allows you to aggregate, visualize, and analyze live data streams.
- Once you send data to ThingSpeak from your devices, you can create instant visualizations of live data without having to write any code
- With MATLAB® analytics inside ThingSpeak, you can write and execute MATLAB code to perform more advanced preprocessing, visualizations, and analyses



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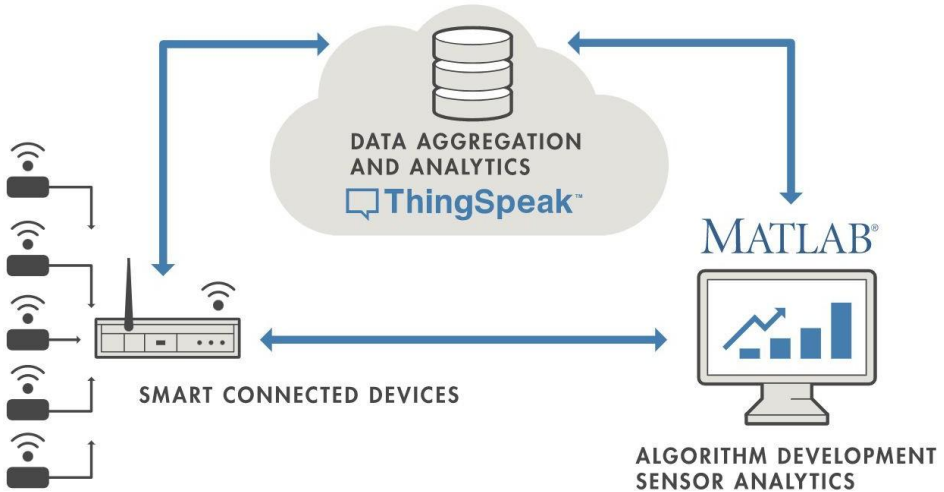
ThingSpeak Features

- Collect data in private channels
- Share data with public channels
- RESTful and MQTT APIs
- MATLAB® analytics and visualizations
- Event scheduling
- Alerts
- App integrations:
  - MATLAB®
  - Arduino®
  - ESP8266 Wifi Module
  - Raspberry Pi™
  - LoRaWAN®



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Setup ThingSpeak Account

- Create a free MathWorks account or sign into ThingSpeak using an existing account (https://thingspeak.com/)
- Select the ThingSpeak channel you want your data to stream into. See Collect Data in a New Channel for help creating a new channel.
  - Record the following for the selected channel:
  - Channel ID, which is listed at the top of the channel view.
  - Write API key, which can be found on the API Keys tab of your channel view.

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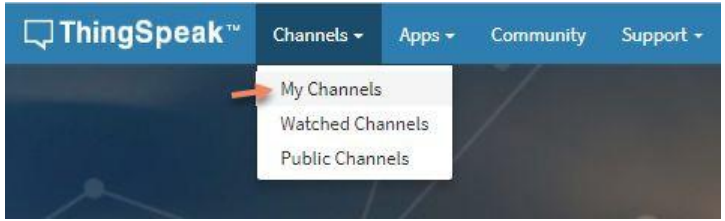
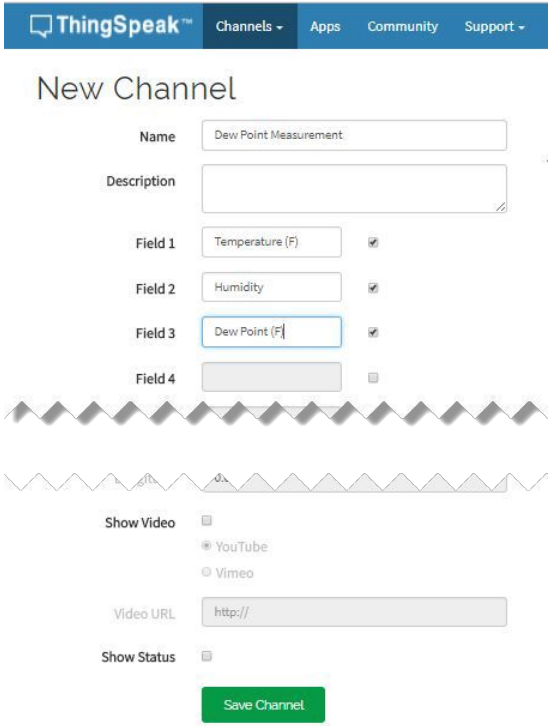
Setup a New Channel in ThingSpeak

- Sign In to ThingSpeak™ using your MathWorks® Account, or create a new MathWorks account.
- Click Channels > MyChannels.



- On the Channels page, click New Channel.
- Check the boxes next to Fields 1–3. Enter these channel setting values:
  - Name: Dew Point Measurement
  - Field 1: Temperature
  - Field 2: Humidity
  - Field 3: Dew Point
- Click Save Channel at the bottom of the settings.



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Simulate Channel Data

- Go to the Apps tab, and click MATLAB Analysis.
- Click New. Select the Custom template, and click Create.
- In the Name field, enter Dew Point Calculation.
- Enter the code below
- Save and Run.

```
% Enter your MATLAB Code below
readChId = 12397;
writeChId = ZZZZZ;       % Replace with your channel number
writeKey = 'XXXXXXXXXXXXXXXX';  % Replace with your channel write key
[temp,time] = thingSpeakRead(readChId,'Fields',4,'NumPoints',20);
humidity = thingSpeakRead(readChId,'Fields',3,'NumPoints',20);
tempC = (5/9)*(temp-32);
b = 17.62;
c = 243.5;
gamma = log(humidity/100) + b*tempC./(c+tempC);
dewPoint = c*gamma./(b-gamma)
dewPointF = (dewPoint*1.8) + 32;
thingSpeakWrite(writeChId,[temp,humidity,dewPointF],'Fields',[1,2,3],... 'TimeStamps',time,'Writekey',writeKey);
```


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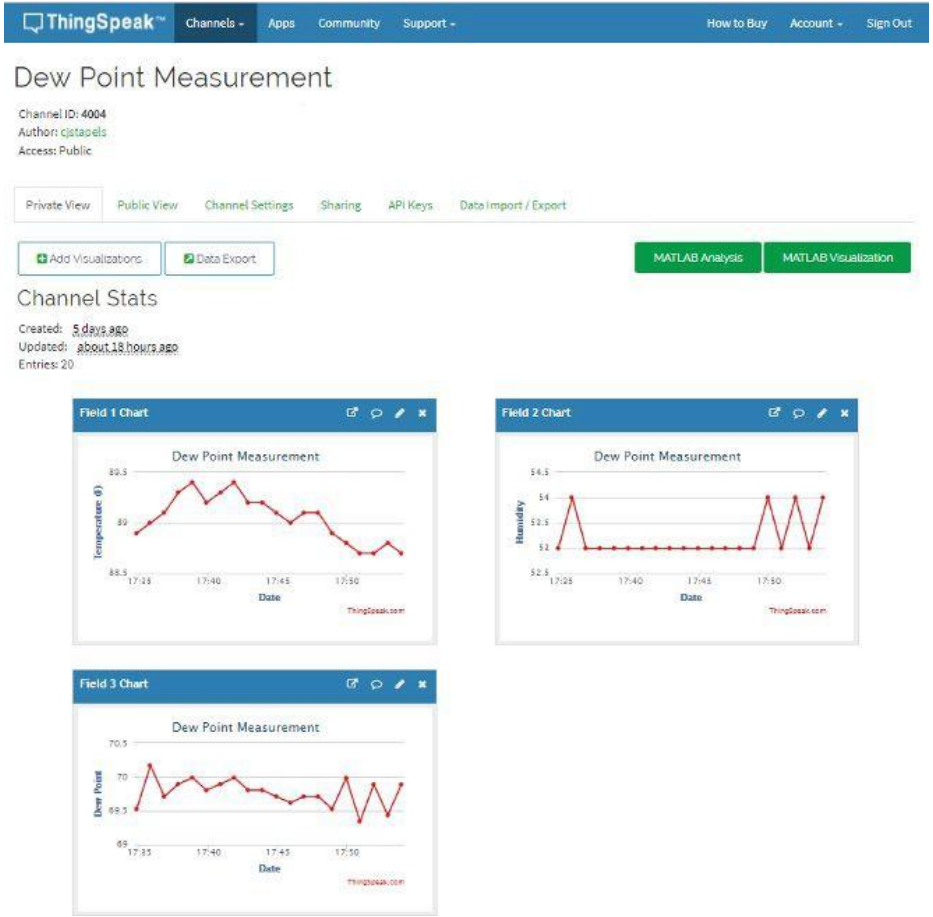
Visualize Channel Data



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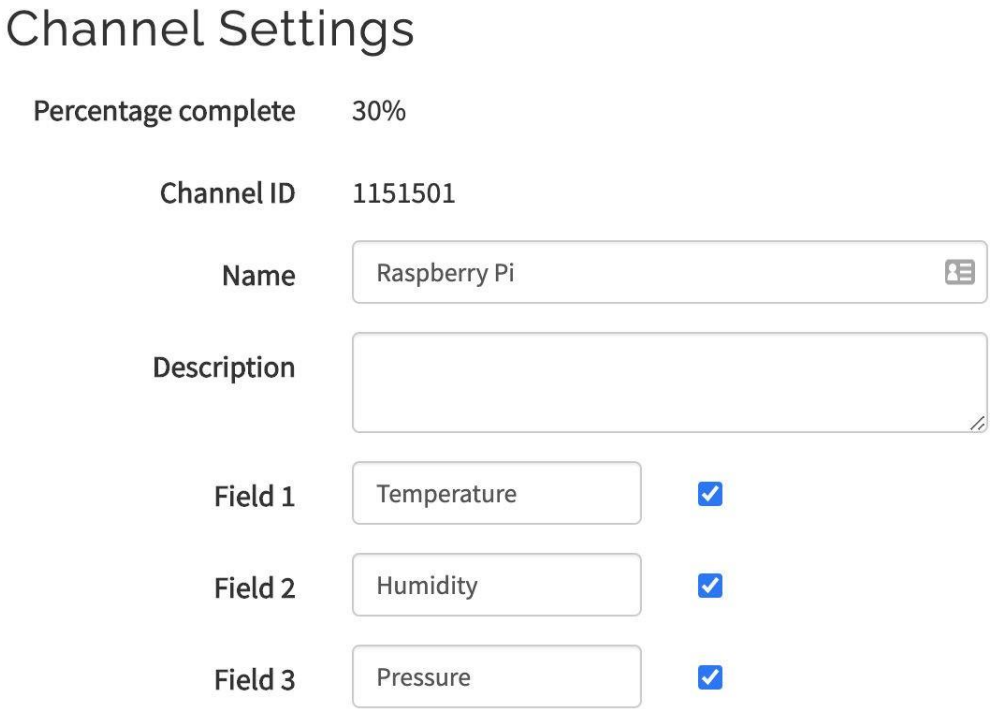
Activity: Setup Channel

- Setup a new channel in ThingSpeak- Raspberry Pi
- Add 3 fields - temperature, humidity and pressure



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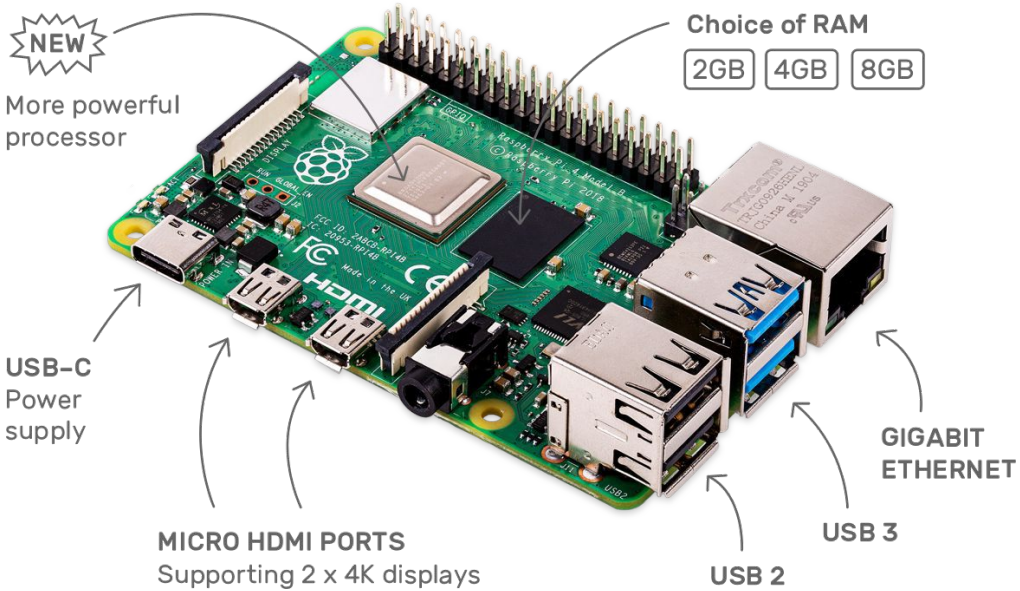
Overview of Raspberry Pi

- The Raspberry Pi is a low cost, credit-card sized computer.
- It enables people of all ages to explore computing, and to learn how to program in languages like Scratch and Python



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Overview of Raspberry Pi

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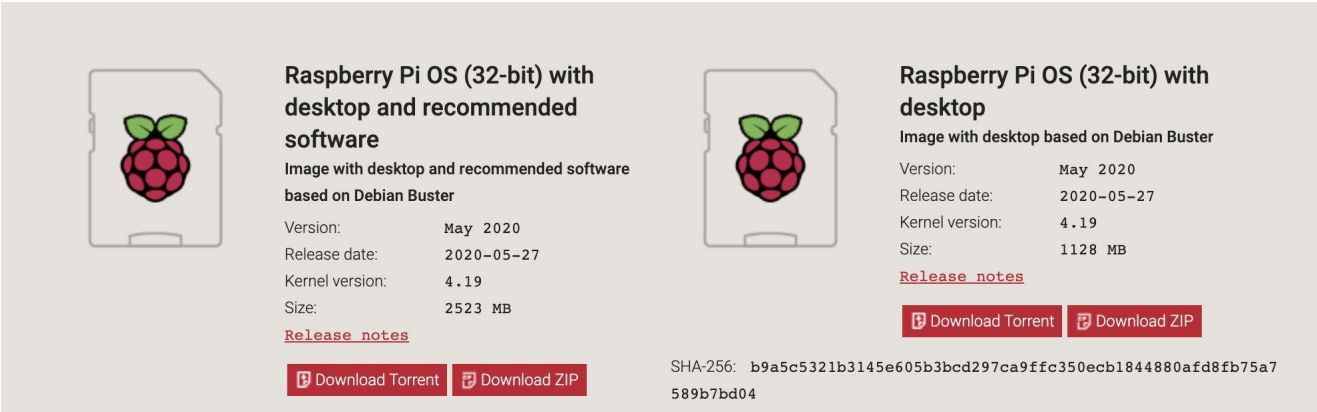
Download Raspberry Pi OS

- Raspberry Pi OS (previously called Raspbian) is the Foundation's official supported operating system. You can install it with NOOBS or download the image
- Raspberry Pi OS comes pre-installed with plenty of software for education, programming and general use. It has Python, Scratch, Sonic Pi, Java and more.
- Goto https://www.raspberrypi.org/downloads/raspberry-pi-os/
- Recommend "Raspberry Pi OS (32-bit) with desktop and recommended software"



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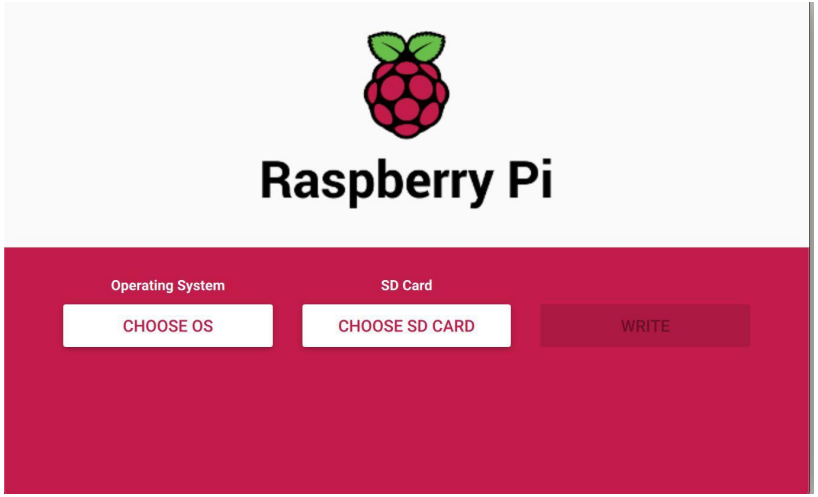
Burn Raspberry Pi Image

- Use Raspberry Pi Imager for an easy way to install Raspberry Pi OS and other operating systems to an SD card ready to use with your Raspberry Pi
- You can get the imager software from https://www.raspberrypi.org/downloads/



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Boot Your New OS

- You can now insert the SD card into the Raspberry Pi and power it up.
- For the official Raspberry Pi OS, the default user name is pi, with password raspberry. You can change the default password straight away to ensure your Raspberry Pi is secure.
- The default keyboard layout is set to UK. Change to US International
- The default timezone is UK. You can change to Singapore.
- Setup your WiFi credential
- Download and the OS update
- Reboot the OS

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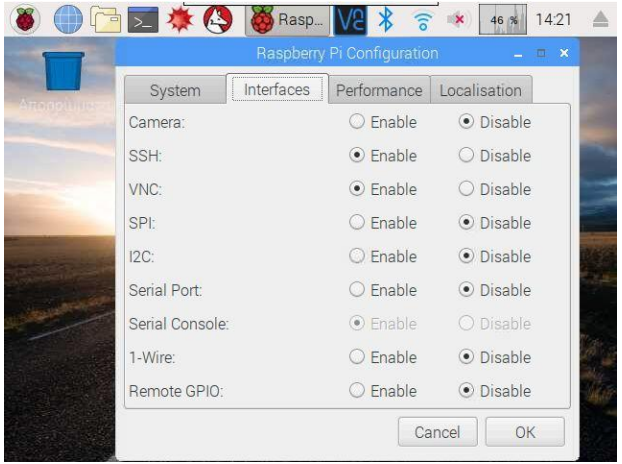
Enable VNC

- If you need to use your Raspberry Pi without connecting it with a monitor (or TV) you need to activate the VNC server service. By using the VNC you will be able to control your RPi from your computer (or laptop)! It's like the TeamViewer but it's only share the desktop of the RPi over a local network.
- The VNC server is already installed in your Raspbian Operating System, you just need to activate it.
- First of all, you need to enable VNC on your Raspberry Pi. Choose Menu-> Preferences-> Raspberry Pi Configuration.



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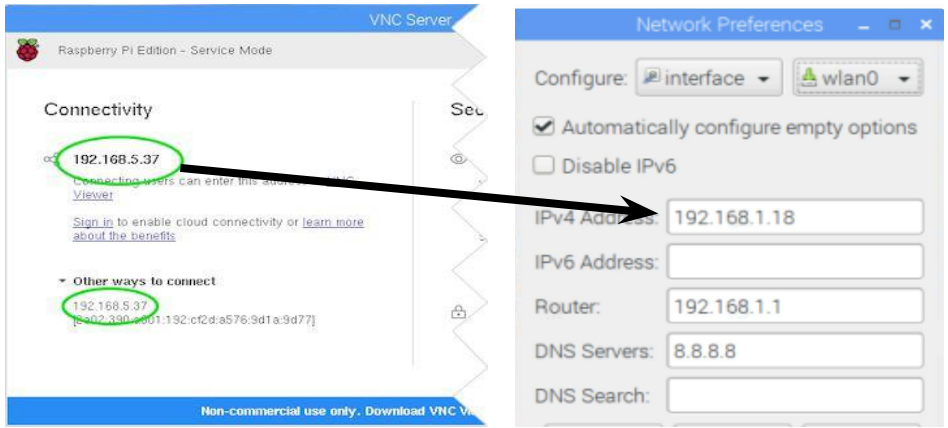
Setup Static IP (Optional)

- Every time that the RPi it's starting (after power on or reboot) it takes a new IP address. For this reason it will be good to set a static IP address. Make a right click on WiFi icon in status bar and select the first choice "Wireless & Wired Network Settings".
- Select wlan0 and write the static IP that you want in "IPv4 Address" field and your router's IP in "Router" field.
- We advise you to use the google dns servers, so in "DNS Servers" field write the address 8.8.8.8.



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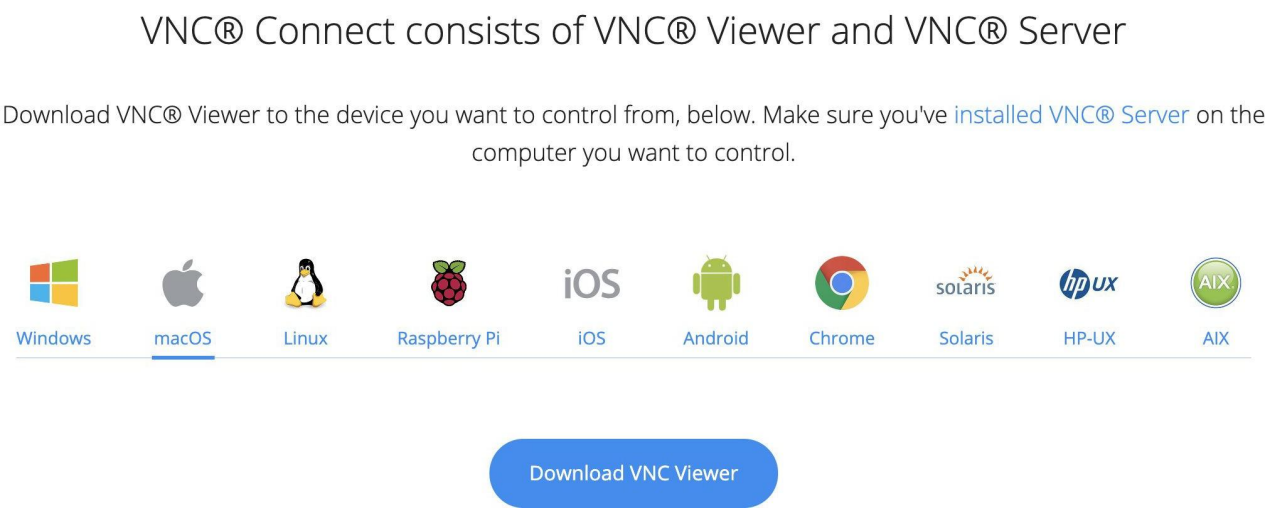
Download VNC Viewer

Download VNC Viewer on your local laptop
https://www.realvnc.com/en/connect/download/viewer/



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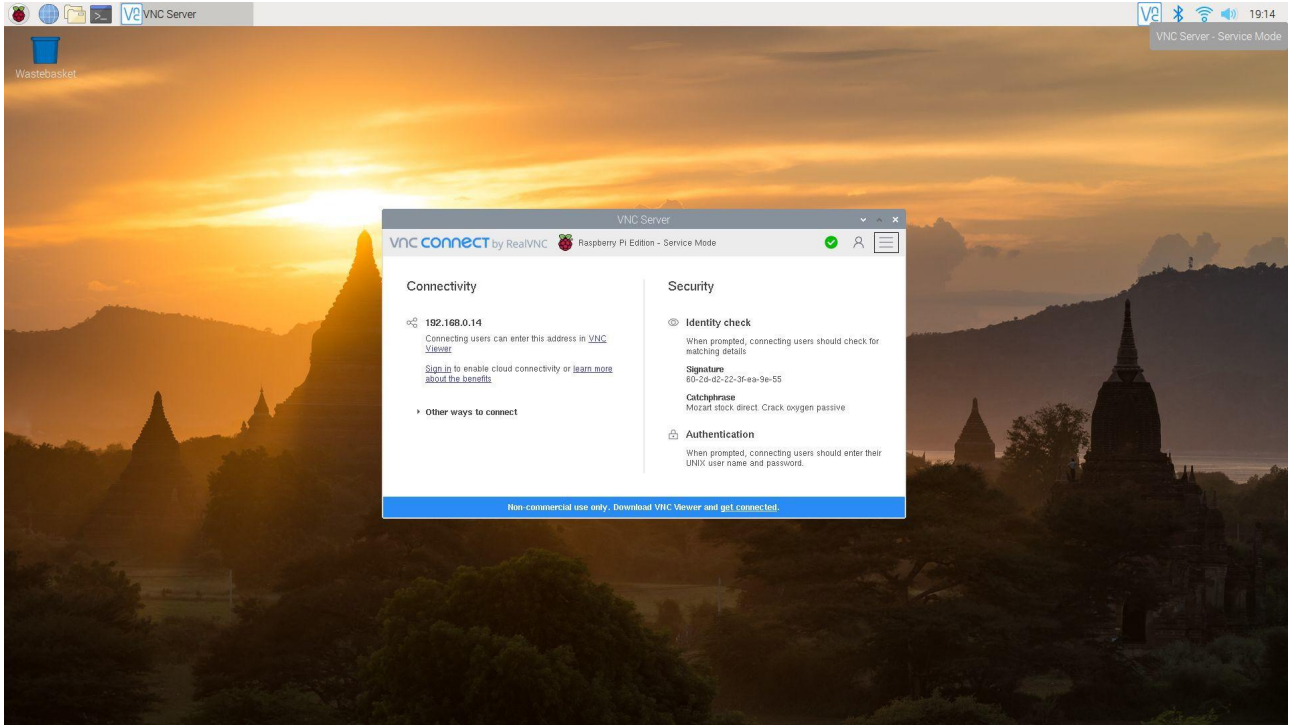
Remote Access Raspberry Pi via VNC

Enter the IP address to your VNC viewer and you can now remote access your Raspberry Pi from laptop



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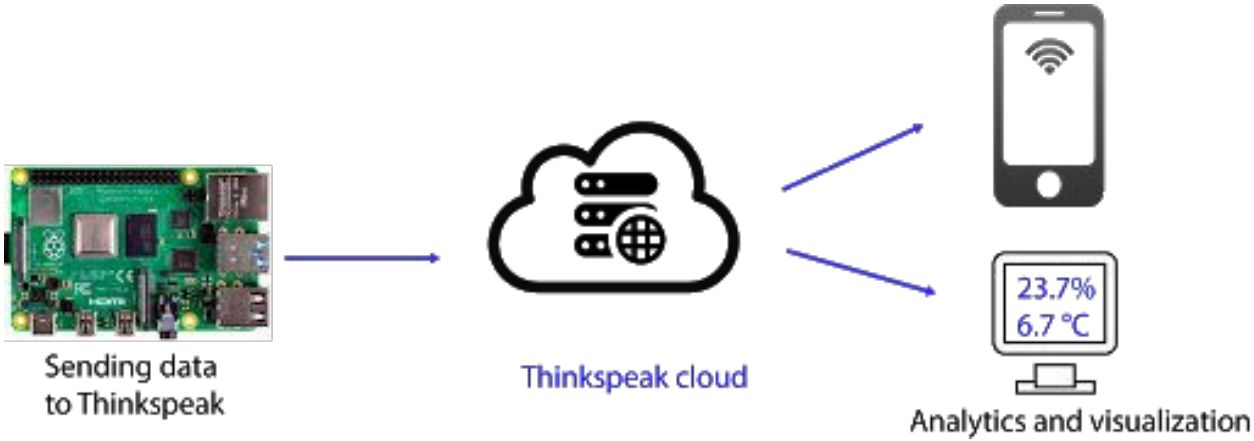
Setup Raspberry Pi IoT

To setup Raspberry Pi IoT

- Collect data from sensors, eg Sense HAT
- Upload the data to cloud eg ThingSpeak
- Perform data analytics and visualization on the cloud
- Display the data on the PC or mobile devices.



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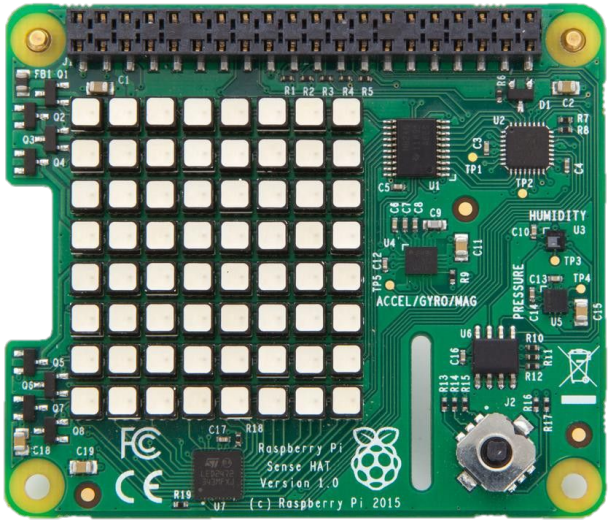
Sense HAT

- The Sense HAT is an add-on board for the Raspberry Pi
- The board allows you to make measurements of temperature, humidity, pressure, and orientation, and to output information using its built-in LED matrix.



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Sense HAT Sensors

- Motion Data
  - Acceleration
  - Gyroscope
  - Orientation
  - Compass
- Environment Data
  - Temperature
  - Humidity
  - Pressure
- Joystick Events
  - up/down/left/right/enter
  - 0: key released- 2: key held down
  - 1: key pressed

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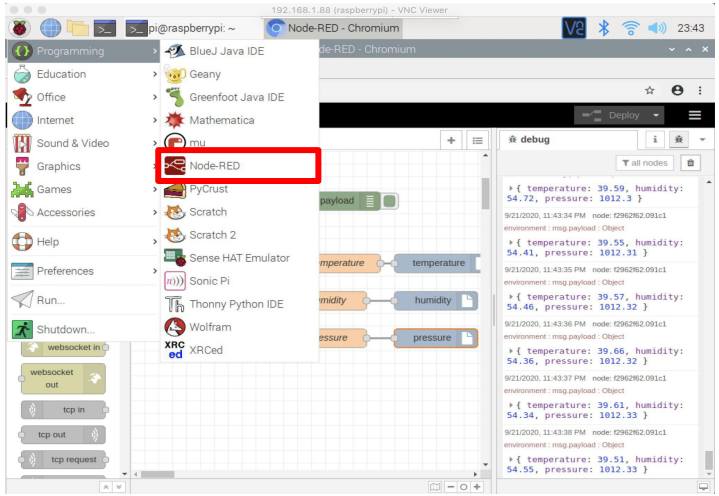
Node-RED

- Node-RED is a visual programming tool for IoT
- It is preinstalled in Raspberry OS
- To start up Node-RED, select Node-RED icon from the dropdown menu
- Open the browser 127.0.0.1:1880 or localhost:1880



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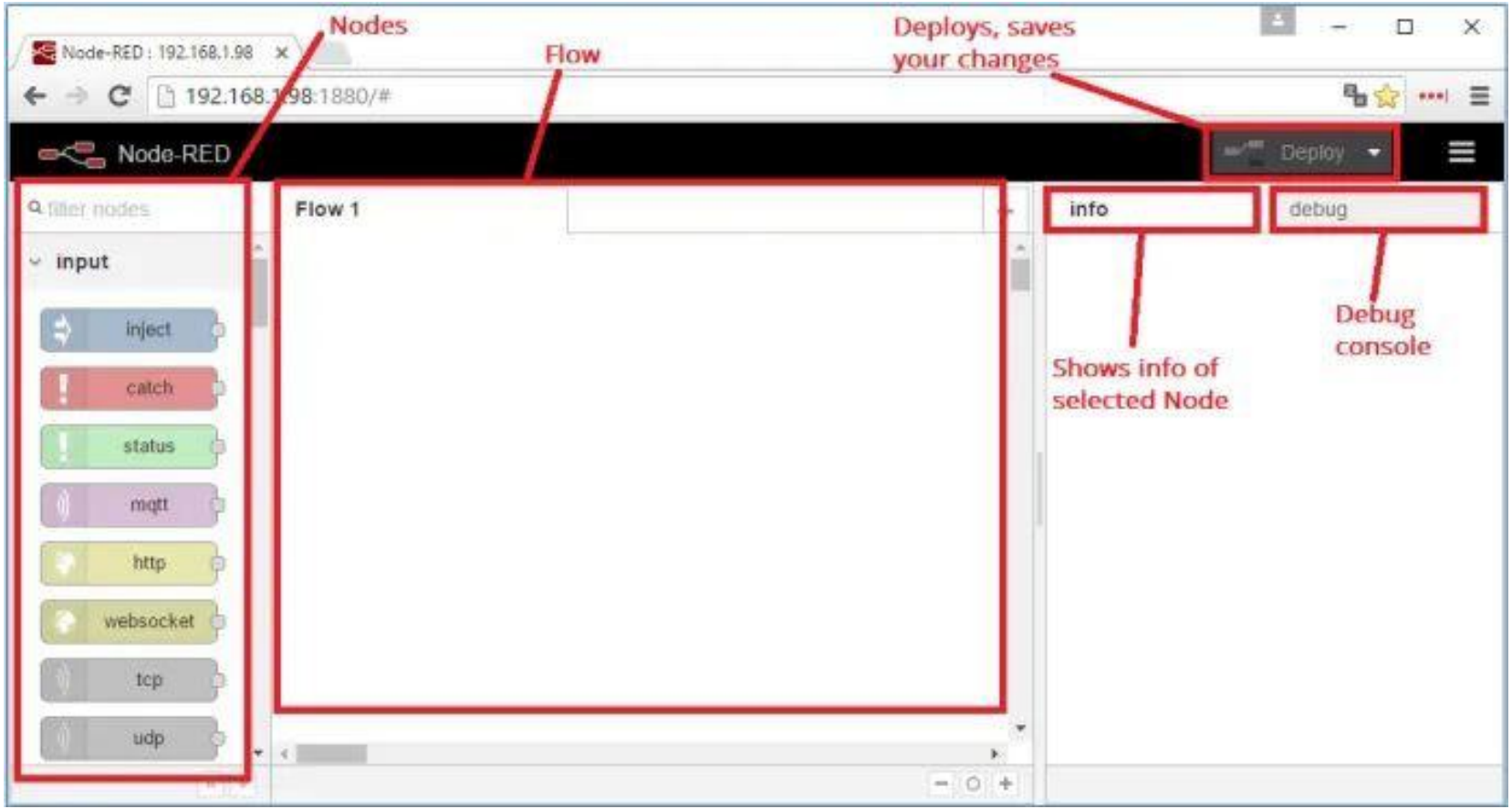
Node-RED Interface



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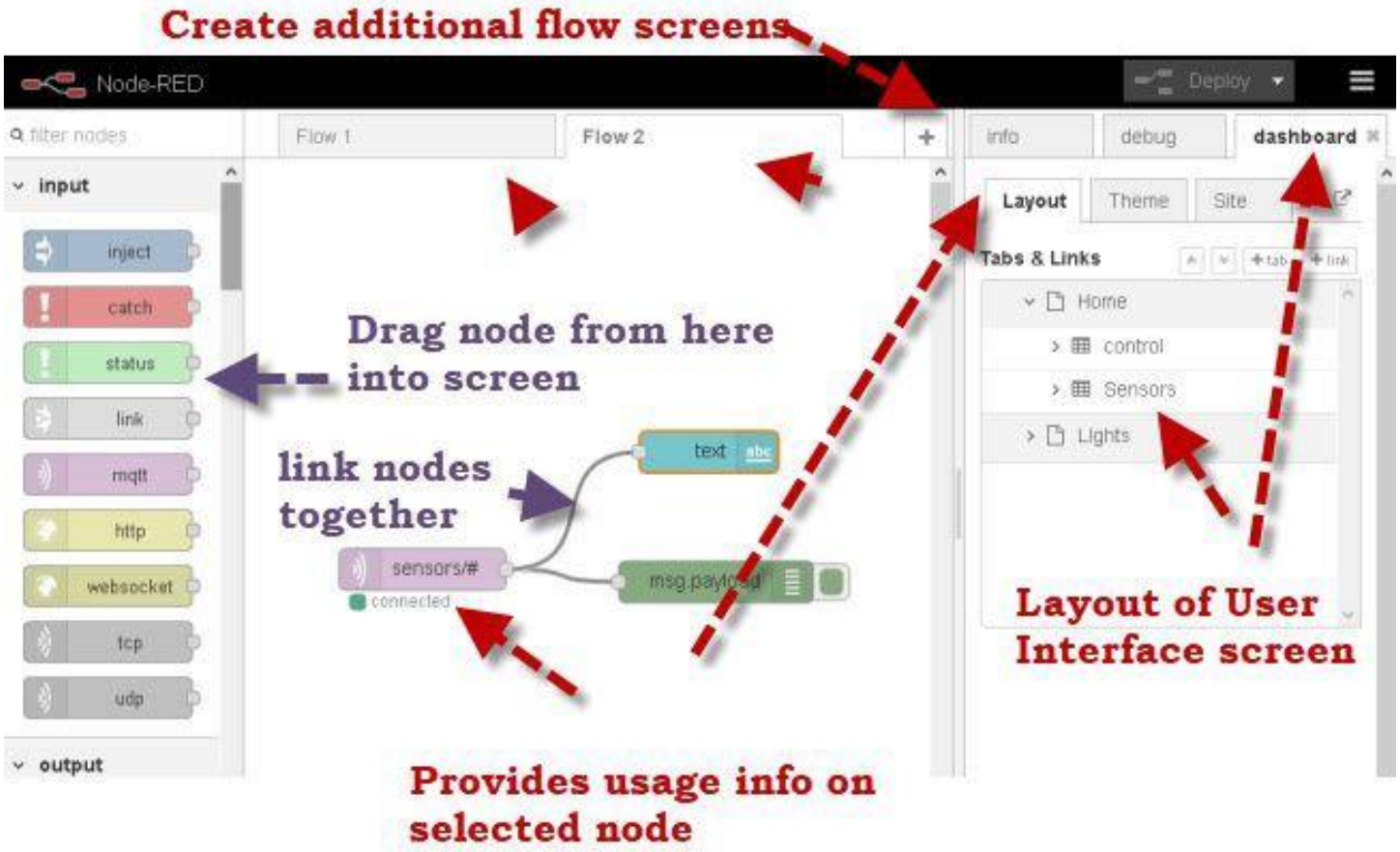
How to Use Node-Red?



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Basic Concepts of Node-RED

Node-RED is based on a graphical interface and rely on three main concepts.

- A project is called a **flow**, and consists of data and functions linked together.
- The **messages** carry data from one node to another.
- The **nodes** are functions that generate, transform or use message

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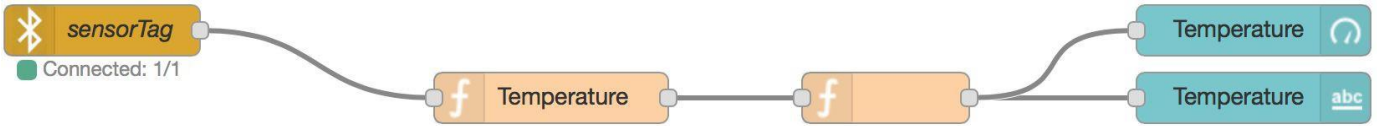
Node-RED Flow

The flow has three parts, from left to right:



On the left, the SensorTag node is the input, connects to the SensorTag CC2650 and acquires the data, here temperature, humidity, pressure and light.

On the middle, the nodes sort the information and process the data

On the right, each node corresponds to an output, here a widget on the dashboard

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Node

- Nodes can be added to the workspace by either:
  - dragging them from the palette,
  - using the quick-add dialog,
  - or importing from the library or clipboard.
- Nodes are joined together by wires via their ports.
- An node can be enabled or disabled using the toggle button at the bottom of the dialog.
- Nodes are wired together by pressing the
- left-mouse button on a node’s port, dragging to the destination node and releasing the mouse button.

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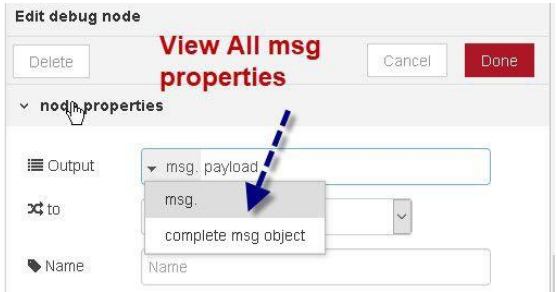
Message Object

- The Message Object is a standard JavaScript object and has several existing properties depending where it originated.
- You can see the message properties by sending the msg to the debug node
- By default the debug node will display the msg.payload property but you can edit the debug node to display all of the message properties: payload, topic and _msgid



Inject Node Message Object

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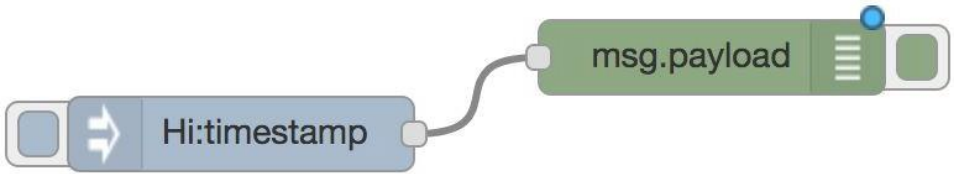
Creating Your First Flow

- Once Node-RED is running, point a local browser at http://localhost:1880.
- The Inject node allows you to inject messages into a flow
- The Debug node causes any message to be displayed in the Debug sidebar.
- Connect the Inject and Debug nodes
- Click the Deploy button



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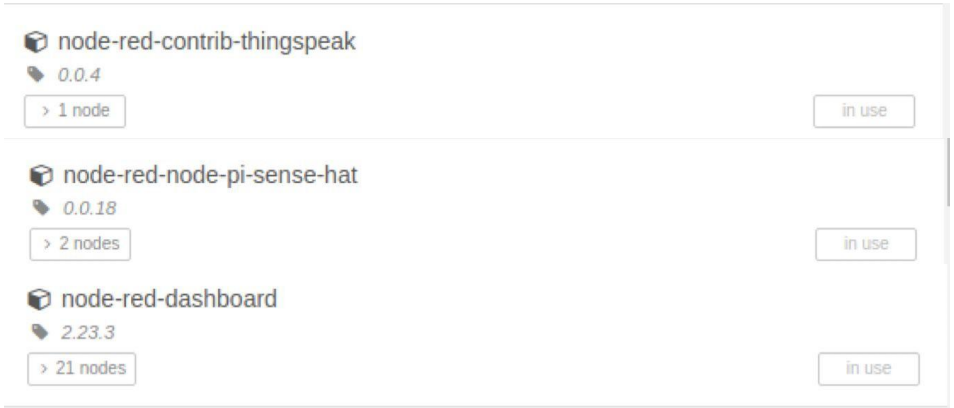
Install New Nodes Using Palette

- Setting -> Mange Palette
- Search for the new node to install
- Install the following 3 nodes
  - node-red-contrib-thingspeak
  - node-red-node-pi-sense-hat
  - node-red-dashboard



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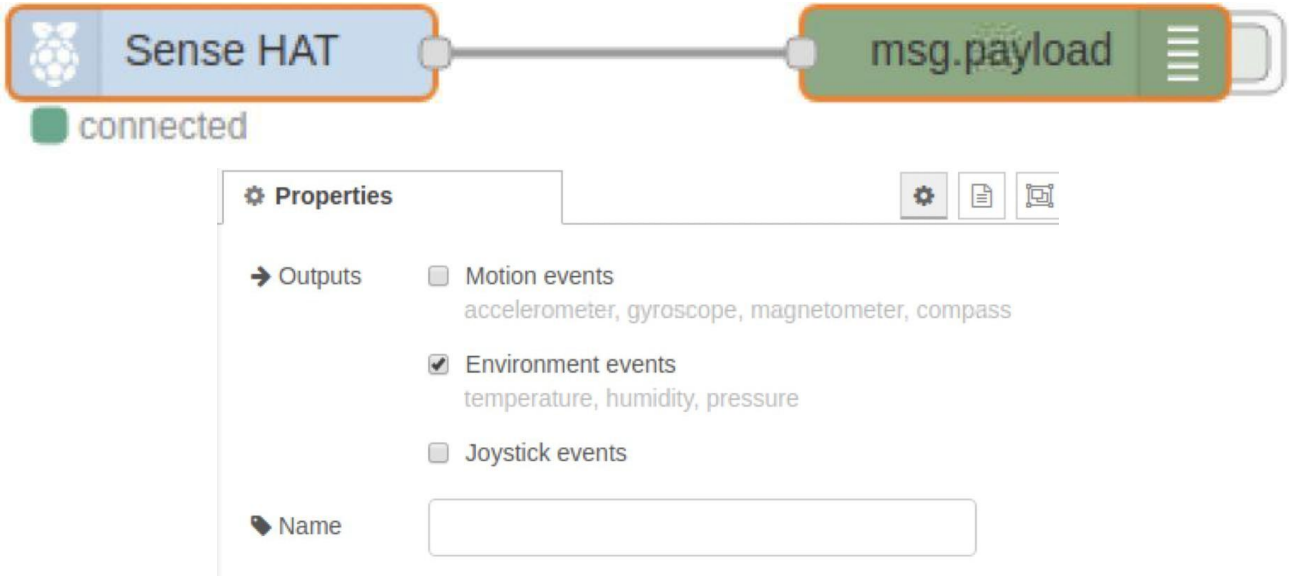
Sensing Environment Data

- Connect the Sense HAT node to the Debug node.
- Select Environment from Sense Hat node setting.
- Deploy the flow



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Function Node

- The Function node allows JavaScript code to be run against the messages that are passed in
- The message is passed in as an object called msg. By convention it will have a msg.payload property containing the body of the message
- Wire the function node between inject and debug nodes
- var newMsg = { payload: msg.payload.length }; return newMsg;

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Sensing Only Temperature Data

- Add a function to display only temperature data

Function

```
1
2  msg.payload= msg.payload.temperature;
3
4  return msg;
5
```


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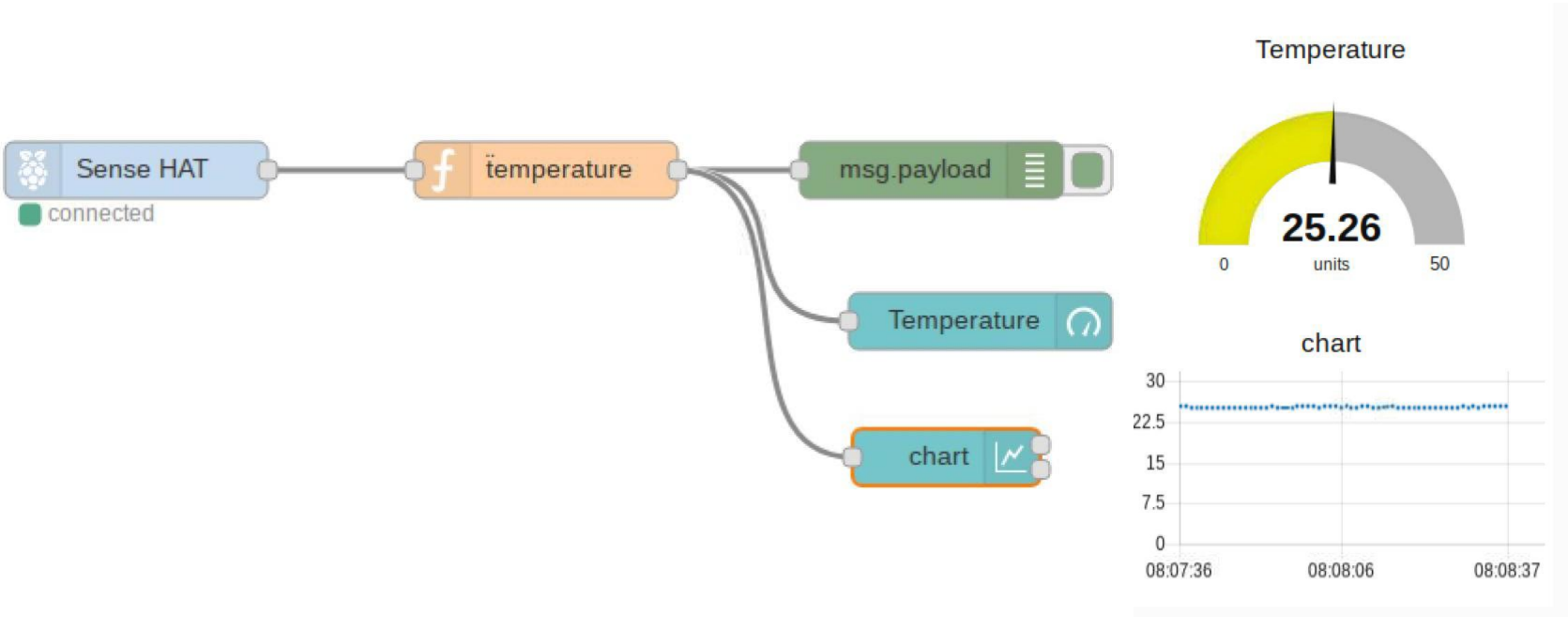
Display Temperature Data

- Add a gauge and chart to show the temperature data
- You can access the UI from http://localhost:1880/ui



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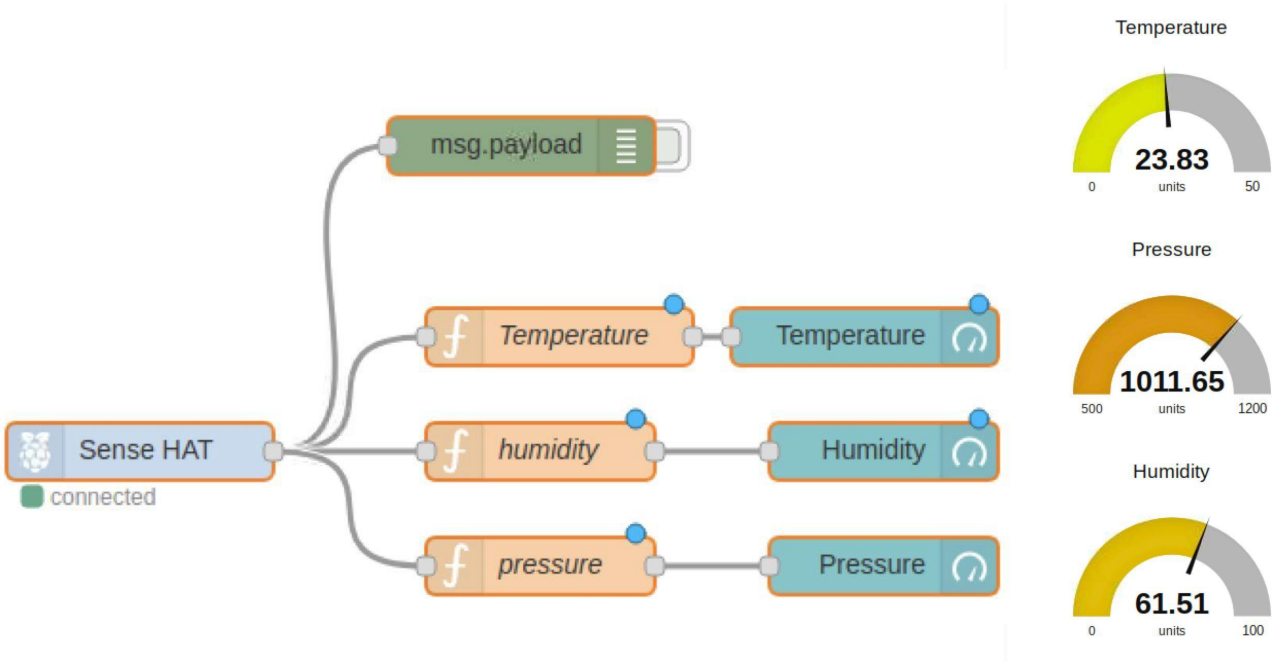
Activity: Display Environmental Data

Add 3 functions to monitor the temperature, humidity and pressure, and display the values using gauges



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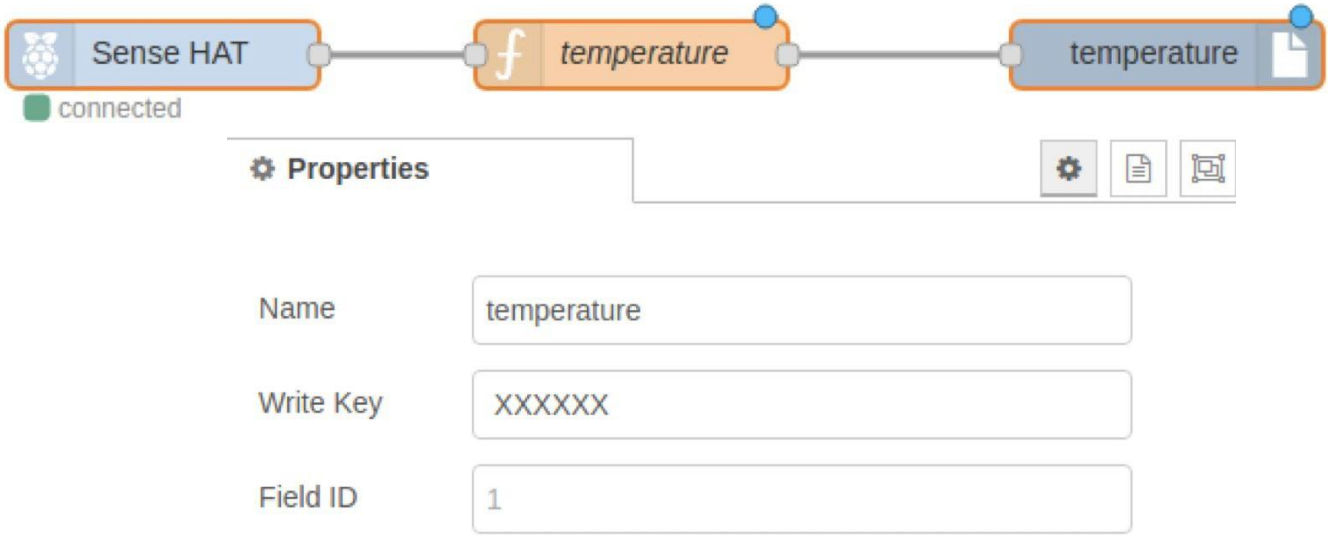
Upload Temperature to Cloud

Connect the temperature data to the ThingSpeak node
Copy the Write API key from ThingSpeak and past the the ThingSpeak node setting.



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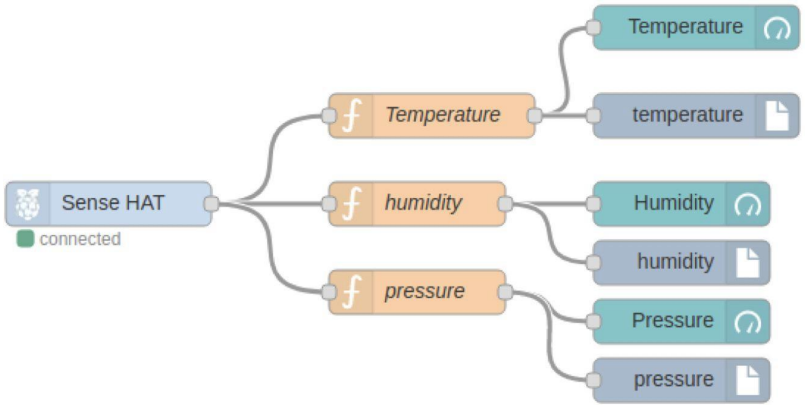
Activity: Upload Environmental Data to Cloud

- Upload the 3 environmental data - temperature, humidity and pressure to the ThingSpeak
- Display the data on the ThingSpeak



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Topic 3
IoT Data Analytics and Visualization

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Data Analytics in the Cloud

- ThingSpeak allows you to use MATLAB to analyze and visualize your data. The analysis and visualization apps provide template code to assist you with basic operations on historic or live data.
- You can use the MATLAB analysis for
  - Explore data collected in a channel or scraped from a website
  - Find and remove bad data
  - Convert data to different units
  - Calculate new data
  - Build data models

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Data Analytics Flow in ThingSpeak

```
%Read data from a channel
readChannelID = xxxxx;
readAPIKey = 'xxxx';
 data = thingSpeakRead(readChannelID, 'ReadKey', readAPIKey);
```

```
%Perform data analysis on the data using Matlab
```

```
 %Write data to a channel
 writeChannelID = xxxxx;
 writeAPIKey = 'xxxxx';
 data = thingSpeakRead(readChannelID, 'ReadKey', readAPIKey);
```


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Sample Code for Data Analytics

```
% TODO - Replace the [] with channel ID to read data from: readChannelID = [];
% TODO - Enter the Read API Key between the '' below: readAPIKey = '';
%% Read Data %%
data = thingSpeakRead(readChannelID, 'ReadKey', readAPIKey);

%% Analyze Data %%
% Add code in this section to analyze data and store the result in the
% 'analyzedData' variable.
analyzedData = data;

% TODO - Replace the [] with channel ID to write data to: writeChannelID = [];
% TODO - Enter the Write API Key between the '' below: writeAPIKey = '';
%% Write Data %%
thingSpeakWrite(writeChannelID, analyzedData, 'WriteKey', writeAPIKey);
```


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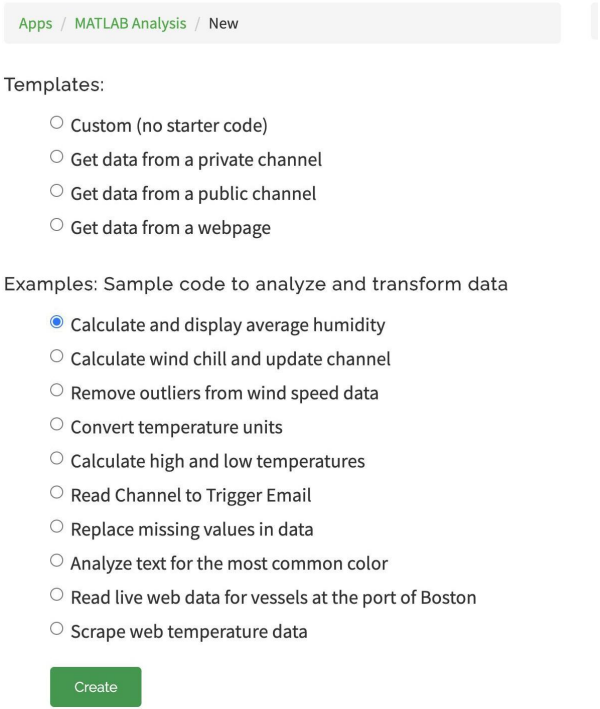
Activity: Review Sample Codes

- Goto Matlab Analysis and Click New
- Review the sample codes to understand more



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Sample Code for Average Humidity

```
readChannelID = 12397;
humidityFieldID = 3;
readAPIKey = '';

% Get humidity data for the last 60 minutes
humidity =
thingSpeakRead(readChannelID,'Fields',humidityFieldID,'NumMinutes',6
0,'ReadKey',readAPIKey);

% Calculate the average humidity
avgHumidity = mean(humidity);
display(avgHumidity,'Average Humidity');

writeChannelID = [];
writeAPIKey = '';
thingSpeakWrite(writeChannelID,avgHumidity,'WriteKey',writeAPIKey);
```


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Sample Code for Average Humidity

```
readChID = 12397;
writeChID = [];
writeAPIKey = '';

% Read air temperature, time stamps, and wind speed from weather station
[temp,time] = thingSpeakRead(readChID,'Fields',4); windSpeed
= thingSpeakRead(readChID,'Fields',2);

% Calculate and display wind chill from weather station data windChill = 35.74
+ (0.6215*temp) - (35.75*windSpeed^0.16) + (0.4275*temp*windSpeed^0.16);
display(windChill,'Wind Chill');

% Plot air temperature, wind speed, and wind chill to your channel
thingSpeakWrite(writeChID,[temp,windSpeed,windChill],'Fields',[1,2,3],...
'TimeStamps',time,'WriteKey',writeAPIKey);
```


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Dew Point Calculation Sample Code

```
% Enter your MATLAB Code below
readChId = 12397;
writeChId = ZZZZZ;     % Replace with your channel number writeKey =
'XXXXXXXXXXXXXXX';  % Replace with your channel write key

[temp,time] = thingSpeakRead(readChId,'Fields',4,'NumPoints',20);
humidity = thingSpeakRead(readChId,'Fields',3,'NumPoints',20); tempC =
(5/9)*(temp-32);
b = 17.62;
c = 243.5;
gamma = log(humidity/100) + b*tempC./(c+tempC);
dewPoint = c*gamma./(b-gamma)
dewPointF = (dewPoint*1.8) + 32;

thingSpeakWrite(writeChId,[temp,humidity,dewPointF],'Fields',[1,2,3],...
'TimeStamps',time,'Writekey',writeKey);
```


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Demo: Max and Min Temperature

- Create a new channel - Max Min Temperature
- Add 2 fields - Max and Min Temperature
- Read the temperature from the Dew Point Calculation Channel
- Analyze the max and min temperature in the last 24 hrs from the Dew Point Calculation Channel
- Write the max and temperature data to the Max Min Temperature channel

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Activity: Max Environmental Data from Raspberry Pi

- Create a new channel - Max Environment Data from Raspberry Pi
- Add 3 fields - Max Temperature, Max Humidity, Max Pressure
- Read the temperature, humidity and pressure from the Raspberry Pi Sense Hat
- Analyze the max environment data in the last 24 hrs
- Write the max environmental data to the Max Environment Data channel from Raspberry Pi

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Data Visualization

- You can use the MATLAB Visualizations app to visualize data in a ThingSpeak™ channel.
- You can view and explore data using interactive visualizations such as an area plot, line plot, or scatter plot in static visualizations using other MATLAB plots.
- You can also make visualizations public and use the URL to embed them on websites.
- ThingSpeak also allows you to use MATLAB to analyze your data.
- The MATLAB Analysis and MATLAB Visualizations apps provide code templates to assist you with basic operations on your historic or live data.

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Data Viz Flow in ThingSpeak

```
%Read data from a channel
readChannelID = xxxxx; fieldID1
= X;
readAPIKey = 'xxxx';
[data, time] = thingSpeakRead(readChannelID, 'Field', fieldID1,
'NumPoints', 30, 'ReadKey', readAPIKey);
```

```
%Perform data visualizatoin on the data using Matlab plot(time,
data);
```


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Sample Code for Line Plot

```
readChannelID = []; fieldID1 = [];

readAPIKey = '';

%% Read Data %%
[data, time] = thingSpeakRead(readChannelID, 'Field', fieldID1,
'NumPoints', 30, 'ReadKey', readAPIKey);



%% Visualize Data %% plot(time, data);
```


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Sample Code for Scatter Plot

```
readChannelID = XXXXXXXX; fieldID1 = X;
fieldID2 = X;

readAPIKey = 'XXXXXX';

% Read first data variable
data1 = thingSpeakRead(readChannelID, 'Field', fieldID1, 'NumPoints', 30, 'ReadKey',
readAPIKey);

% Read second data variable
data2 = thingSpeakRead(readChannelID, 'Field', fieldID2, 'NumPoints', 30, 'ReadKey',
readAPIKey);

%% Visualize Data %% scatter(data1, data2);
```


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Sample Code for Temperature and Humidity Scatter Plot

```
readChannelID = 12397;
TemperatureFieldID = 4;
HumidityFieldID = 3;

readAPIKey = '';

data = thingSpeakRead(readChannelID,'Fields',[TemperatureFieldID HumidityFieldID],
'NumPoints',300, 'ReadKey',readAPIKey);

temperatureData = data(:,1);
humidityData = data(:,2);

% Visualize the data
scatter(temperatureData,humidityData);
xlabel('Temperature');
ylabel('Humidity');
```


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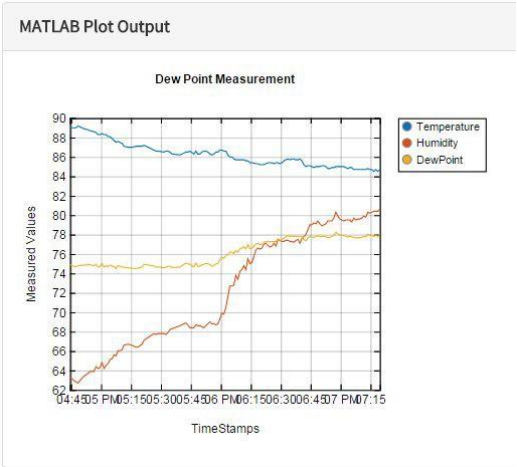
Sample Code for Dew Point

```
readChId = ZZZZZ      % Your Channel ID readKey = 'XXXXXXXXXXX' %Your Read API Key
[dewPointData,timeStamps] = thingSpeakRead(readChId,'fields',[1,2,3],...
'NumPoints',100,'ReadKey',readKey); plot(timeStamps,dewPointData);
xlabel('TimeStamps'); ylabel('Measured Values'); title('Dew Point Measurement');
legend({'Temperature','Humidity','Dew Point'}); grid on;
```



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Activity: Max Environmental Data from Raspberry Pi

- Create a new channel - Max Environment Data from Raspberry Pi
- Add 3 fields - Max Temperature, Max Humidity, Max Pressure
- Read the temperature, humidity and pressure from the Raspberry Pi Sense Hat
- Analyze the max environment data in the last 24 hrs
- Write the max environmental data to the Max Environment Data channel from Raspberry Pi

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Topic 4
Monitor Data to Trigger Control from Cloud

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IoT Communication Standards

- IoT Architecture: The IoT Architecture provides a blueprint for the design and implementation of IoT systems. It defines the basic building blocks of an IoT system and the interfaces between them.
- Security Guidelines: Security is a critical aspect of IoT systems, and there are various guidelines and best practices that can help ensure the security of IoT devices and networks. These guidelines include secure boot, secure communication, and device authentication.
- Connectivity Standards: There are several connectivity standards for IoT, including Wi-Fi, Bluetooth, Zigbee, Z-Wave, LoRaWAN, Sigfox, and NB-IoT. Each of these standards has its own unique features and characteristics that make them suitable for different IoT applications.

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IoT Communication Standards

- Data Communication Standards: IoT devices generate large amounts of data, and there are various communication standards that can be used to transmit this data. These standards include MQTT, CoAP, REST, and WebSocket.
- Interoperability Guidelines: Interoperability is important for IoT devices to work together seamlessly. There are several guidelines and standards that can help ensure interoperability, such as the Open Connectivity Foundation (OCF), the Zigbee Alliance, and the Thread Group.

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IoT Communication Standards

- Energy Efficiency Guidelines: IoT devices often operate on battery power or other low-power sources, which requires energy-efficient design and management. Guidelines for energy efficiency in IoT include minimizing power consumption, optimizing transmission schedules, and using low-power communication technologies.
- Cloud Communication Standards: Cloud computing is an important aspect of many IoT systems, and there are various communication standards that can be used to transmit data between IoT devices and the cloud. These standards include HTTP, AMQP, and MQTT.

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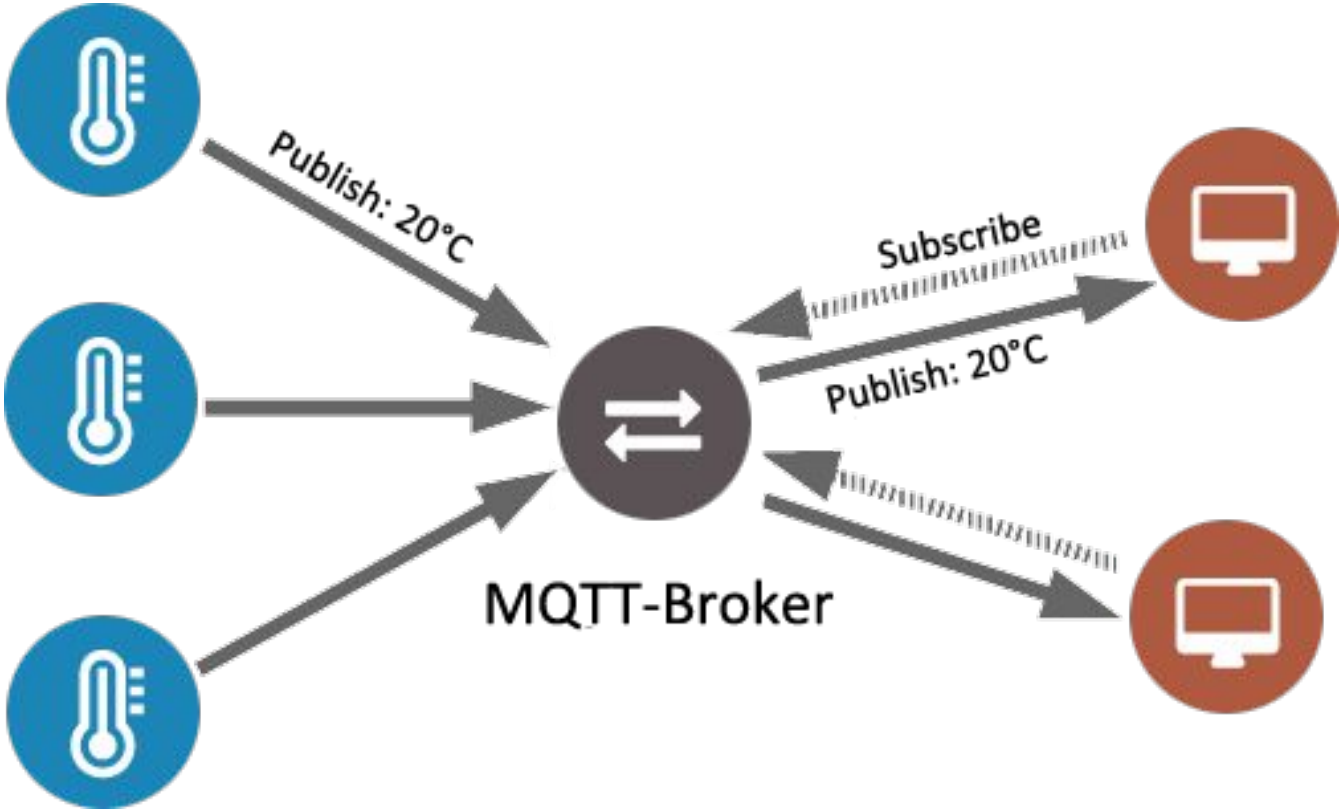
MQTT Standard



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MQTT Standard

MQTT is a publish/subscribe architecture that is developed primarily to connect bandwidth and power-constrained devices over wireless networks.
It is a simple and lightweight protocol that runs over TCP/IP sockets or WebSockets.
MQTT over WebSockets can be secured with SSL.
The publish/subscribe architecture enables messages to be pushed to the client devices without the device needing to continuously poll the server.
The MQTT broker is the central point of communication, and it is in charge of dispatching all messages between the senders and the rightful receivers.
A client is any device that connects to the broker and can publish or subscribe to topics to access the information. A topic contains the routing information for the broker.
Each client that wants to send messages publishes them to a certain topic, and each client that wants to receive messages subscribes to a certain topic.
The broker delivers all messages with the matching topic to the appropriate clients.

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Key Features of MQTT

- Pub/Sub Model: MQTT uses a publish/subscribe (pub/sub) messaging model, where devices can publish messages to a topic, and other devices can subscribe to that topic to receive those messages.
- QoS Levels: MQTT supports three different quality of service (QoS) levels: 0, 1, and 2. QoS 0 delivers messages with no guarantee of delivery, while QoS 1 and QoS 2 provide increasing levels of reliability.
- Lightweight: MQTT is designed to be lightweight and efficient, which makes it suitable for use in resource-constrained devices, such as sensors and microcontrollers. The protocol header is only 2 bytes long, and messages can be as small as a single byte

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Key Features of MQTT

- TCP/IP Based: MQTT uses TCP/IP as its underlying transport protocol, which provides reliable, connection-oriented communication between devices.
- Last Will and Testament: MQTT provides a Last Will and Testament (LWT) feature, which allows devices to specify a message that will be sent to a topic if the device unexpectedly disconnects from the network.
- Retained Messages: MQTT supports retained messages, which are messages that are stored on the MQTT broker and sent to any new subscribers that connect to the topic.
- Security: MQTT provides security features, such as authentication and encryption, to protect the privacy and integrity of IoT data.

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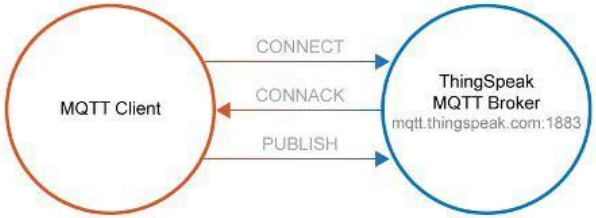
How ThingSpeak MQTT Work

- ThingSpeak™ has an MQTT broker at the URL mqtt.thingspeak.com and port 1883.
- MQTT Publish - The Write API Key is required to publish. The broker acknowledges a correct CONNECT request with CONNACK.



1. Publish to a channel feed

channels/<channelID>/publish/<writeAPIKey>

2. Publish to a channel feed

channels/<channelID/publish/fields/field<fieldnumber>/<writeAPIKey>

1. Subscribe to a channel feed

channels/<channelID>/subscribe/<format>/<api_key>

2. Subscribe to a private channel feed

channels/<channelID>/subscribe/fields/field<fieldNumber>/<apiKey>

3. Subscribe to all fields of a channel

channels/<channelID>/subscribe/fields/+/<apiKey>

<api_key> is not required to subscribe to public channels

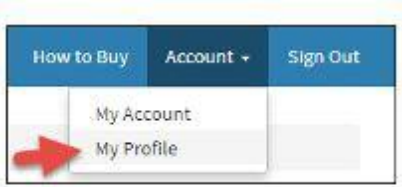
- MQTT Subscribe - To subscribe to a public or private channel, you must also supply a user name and your MQTT API key as password when you connect. The broker acknowledges a correct SUBSCRIBE request with SUBACK

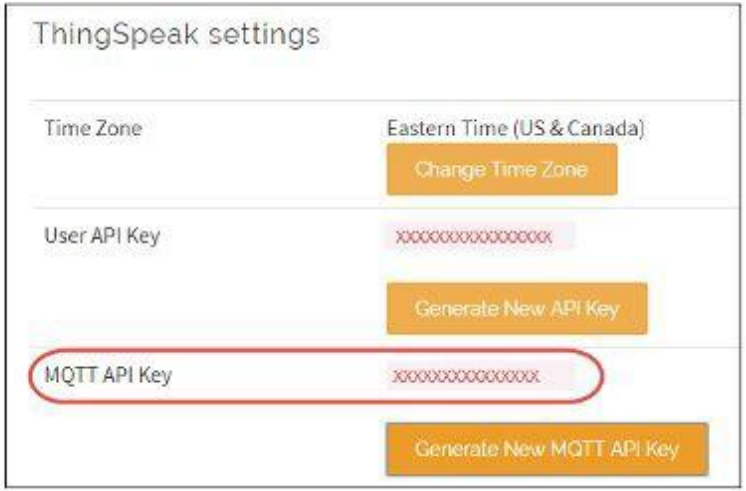
ThingSpeak Subscribe MQTT Info

- Subscribe to a Channel Field Feed
  - channels/<channelID>/subscribe/<format>
  - channels/<channelID>/subscribe/<format>/<apiKey>
  - channels/<channelID>/subscribe/fields/field<fieldnumber>
  - channels/<channelID>/subscribe/fields/field<fieldnumber>/<api key>
  - channels/<channelID>/subscribe/fields/+
  - channels/<channelID>/subscribe/fields/+/<apikey>
- Note:
  - The format is either json, csv or xml
  - ThingSpeak™ MQTT requires an MQTT API Key to subscribe to channel updates for both private and public channels.
  - Supply any unique user name and your MQTT API Key as a password parameter to the ThingSpeak MQTT broker when connecting.
  - Choose Account > My Profile to see your MQTT API key. ThingSpeak MQTT supports only QoS 0.

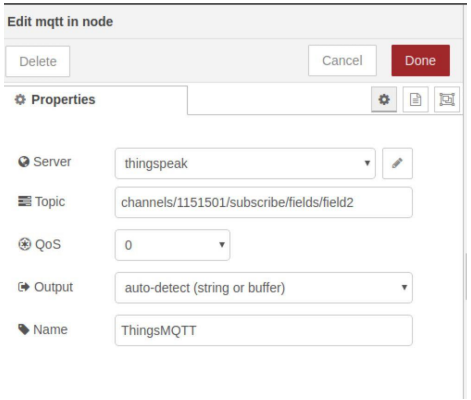
MQTT API Key

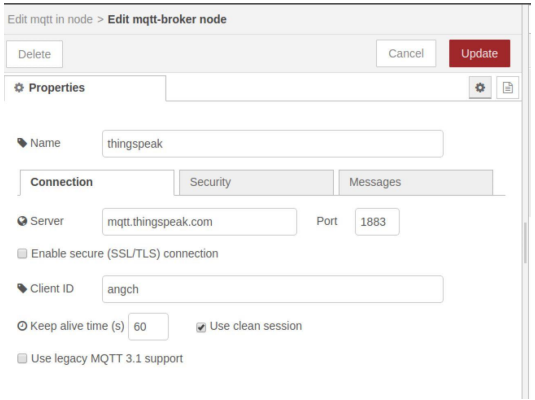
- To use MQTT subscribe, supply any unique user name and the MQTT API Key as your password when you connect.
- To see your MQTT API key, choose Account > My Profile.

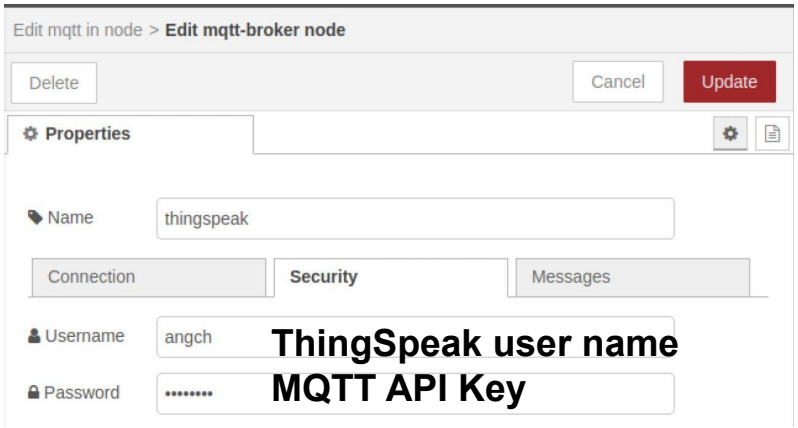




Configure MQTT node on Node-Red

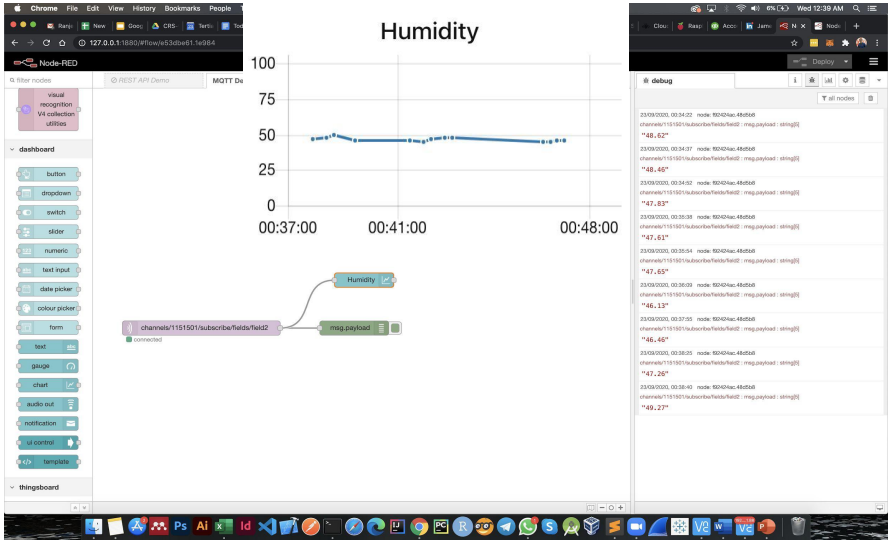






Activity: Read Environmental Data from the Cloud

- Download and install Node-RED on your PC
- Install the Dashboard node
- Read the humidity data from ThingSpeak via MQTT API method
- Display the data on Node-red dashboard



Install Node-RED Locally

- Before you can install Node-RED, you must have a working install of Node.js. Download and install Node.js on https://nodejs.org/en/
- Node-Red can be installed on Windows, Mac OS and Linux. Follow the instructions on https://nodered.org/docs/getting-started/installation

- After you install node.js, enter the following command to install node-red on your Window Command Prompt or Mac terminal
- npm install -g --unsafe-perm node-red

React

- React works with ThingHTTP, ThingTweet, and MATLAB® Analysis apps to perform actions when channel data meets a certain condition. For example, you can have a mobile app report your latitude and longitude to a ThingSpeak™ channel. When your position is within a certain distance of your house, have ThingHTTP turn on your living room lights.
- Define Reactions to Channel Data
  - Select Apps > React.
    - Click New React
    - Select:
    - Condition Type
    - Test Frequency
    - Action
    - Options
  - Click Save React.
- Condition Types:
- ThingSpeak channels support numeric sensor data, text, strings, status updates, and geo location information. Use these condition types to specify conditions that trigger reactions based on your channel data.

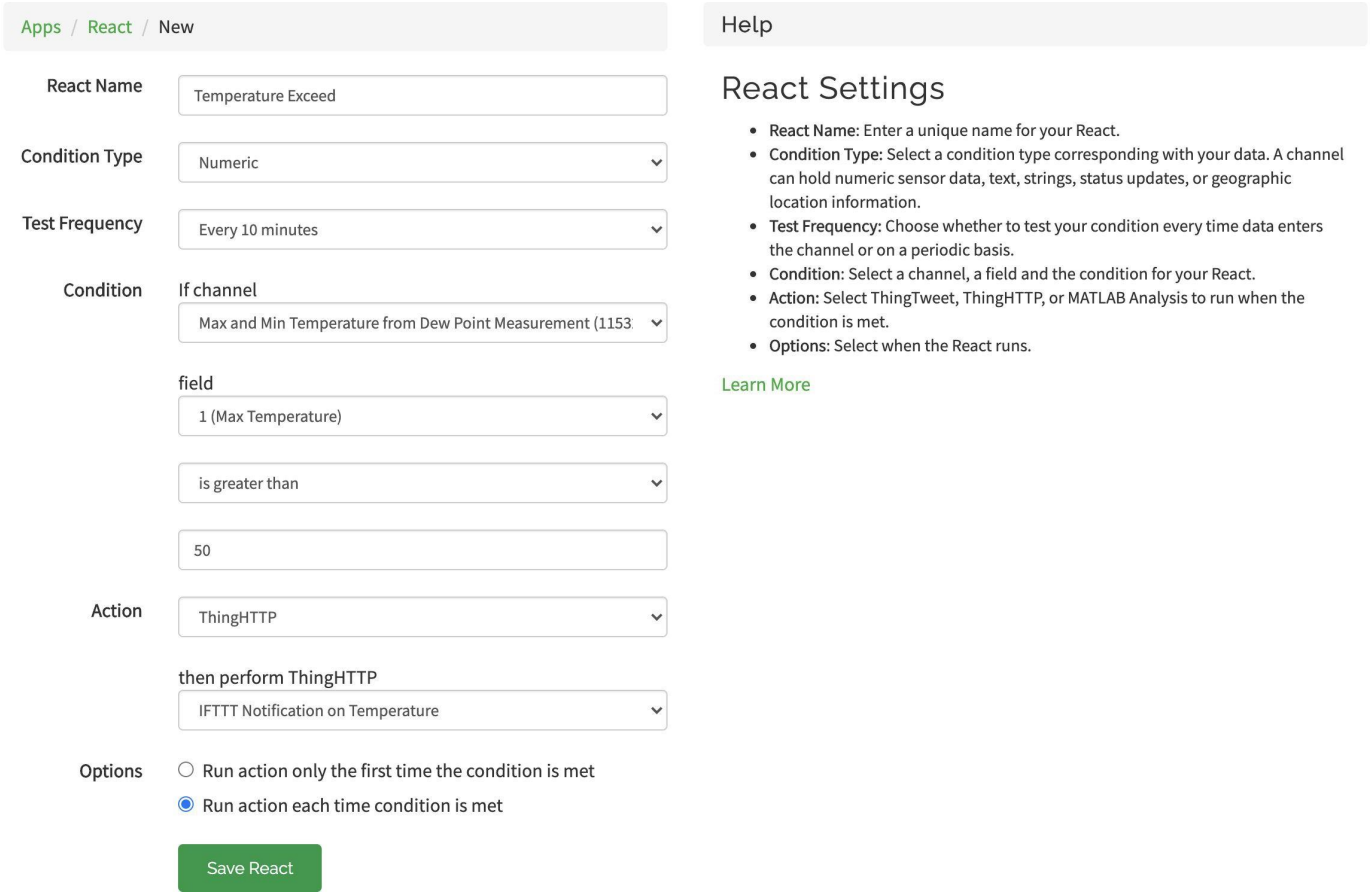
Action on Data

- Use ThingSpeak™ apps to trigger an action or transform and visualize data
- You can upload data from the web or send data from devices to a ThingSpeak IoT platform channel. Use these apps to transform and visualize data on ThingSpeak channels or trigger an action.
- Use the Alerts API interface to schedule emails in response to ThingSpeak data.
- Use the TalkBack API interface to control your TalkBack device queue.

ThingHTTP

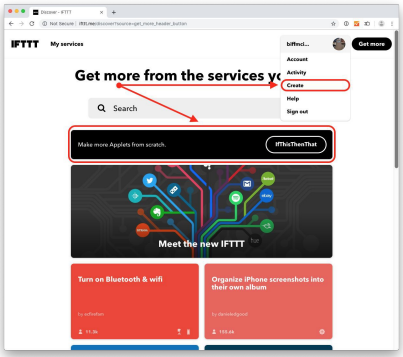
- ThingHTTP enables communication among devices, websites, and web services without having to implement the protocol on the device level.
- You specify actions in ThingHTTP, which you trigger using other ThingSpeak™ apps such as TweetControl, TimeControl, and React.
- To create a ThingHTTP Request
- Create a new ThingHTTP request. Click Apps > ThingHTTP, and then click New ThingHTTP.
- Specify settings. Not all settings are required for each type of request.
- Save the request. Your ThingHTTP is available for future use by clicking Apps and choosing ThingHTTP.

Example of React Setting



Setup Email Alert via IFTTT

- IFTTT is a web service that lets you create applets that act in response to another action. You can use the IFTTT Webhooks service to create web requests to trigger an action.
- You can create your own Applet via ifttt.com/create.
- You can also access this by visiting the profile icon in the top right of ifttt.com and clicking Create.

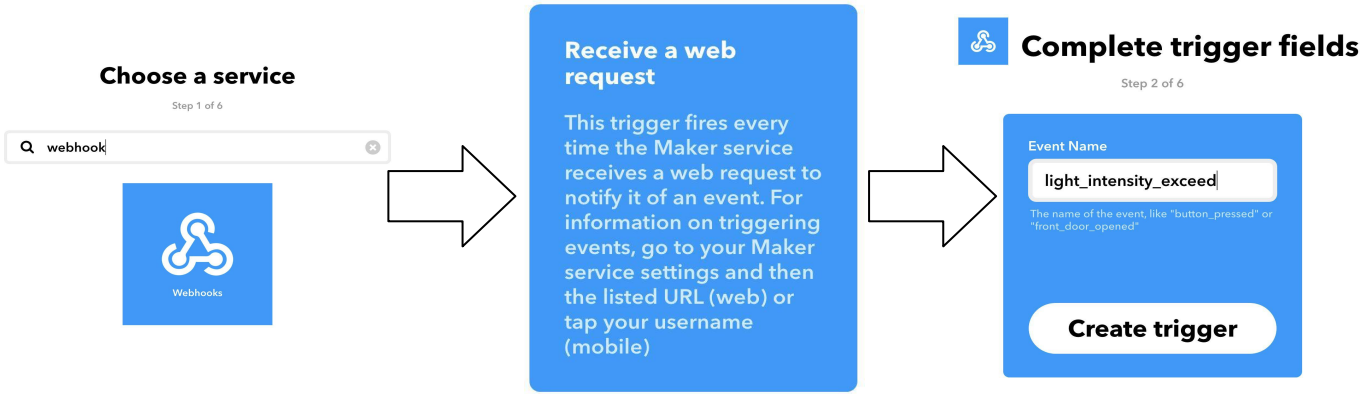


Step 1 - Create Trigger

Click on the + icon to search for a trigger

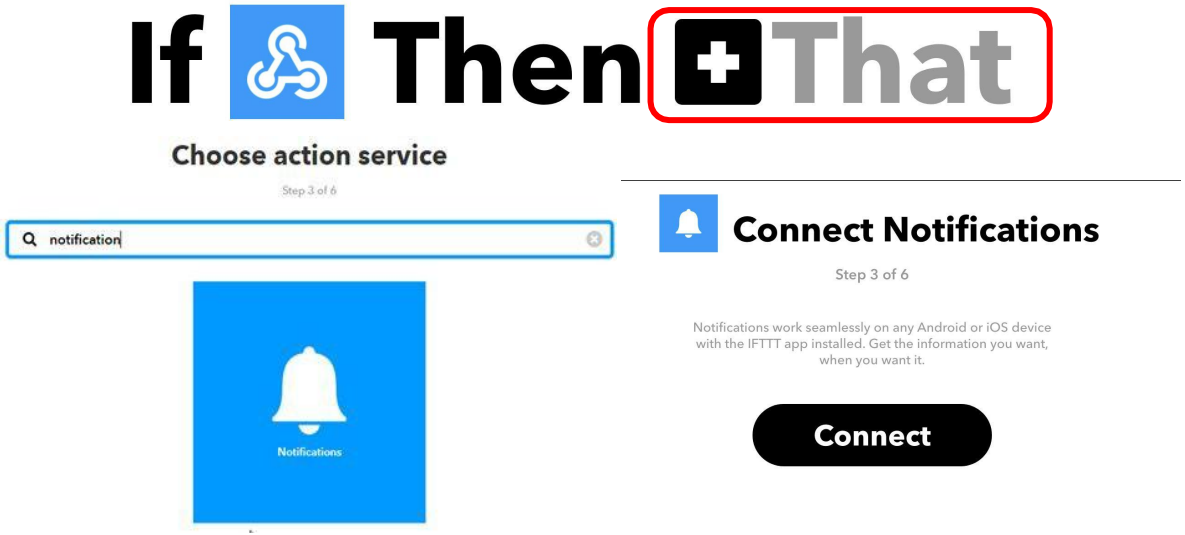
Step 2 - Choose Trigger

- Select the Webhooks service. Enter Webhooks in the search field.
- Select the Webhooks card.
- After you select Webhooks as the trigger, click the Receive a web request card to continue.
- Complete the trigger fields. Enter an event name.



Step 3 - Create Action

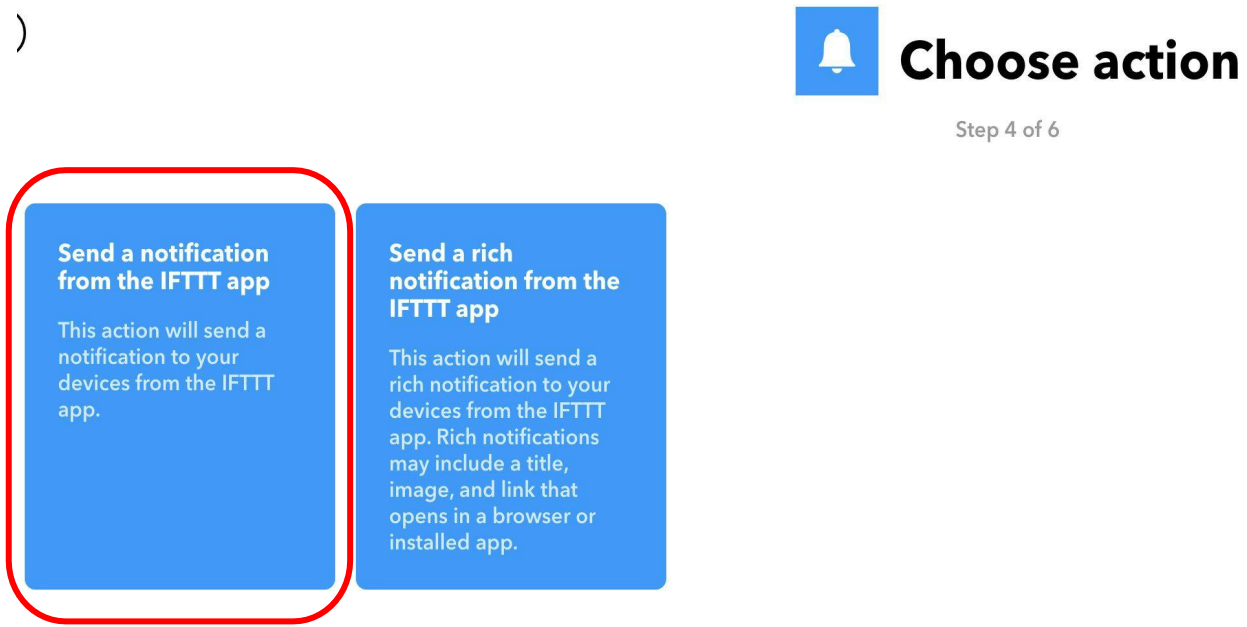
- Select the resulting action. Click the word that. The trigger word this is now the Webhooks icon.
- Enter notification in the search bar, and click the Notifications box..



Step 4 - Choose Action

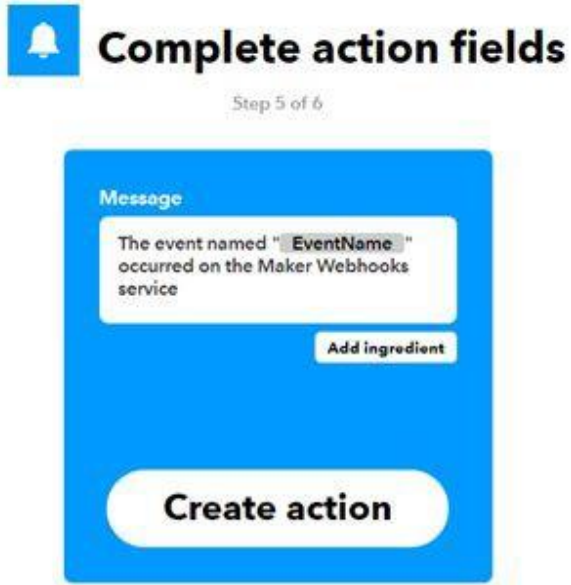
Then select Send a notification from the IFTTT app

)



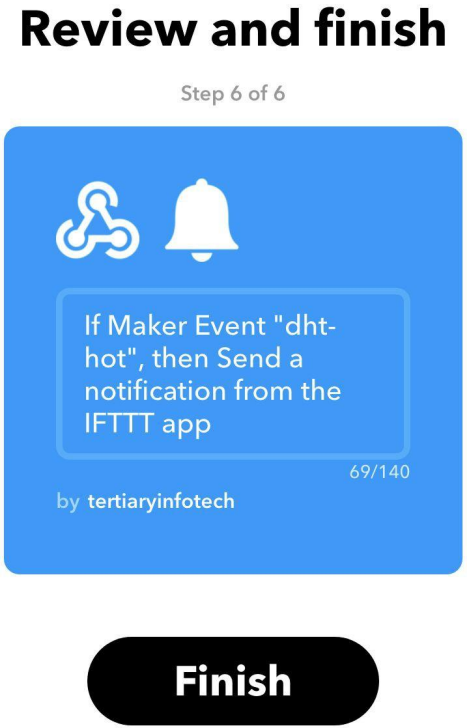
Step 5 - Complete Action Fields

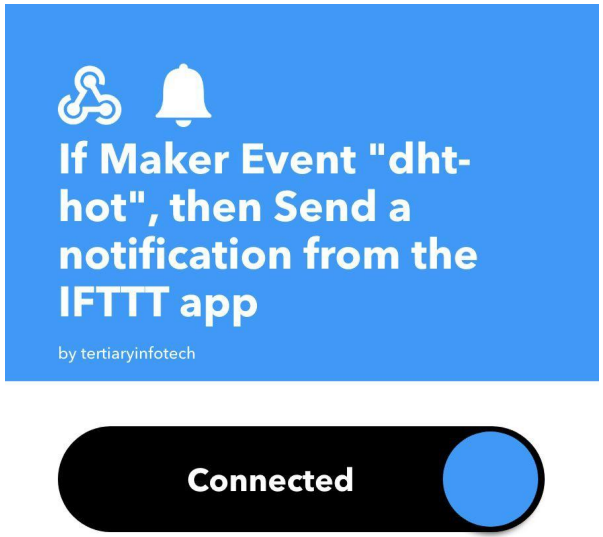
- Click Create action to finish the new applet.
- IFTTT will pass this name of the event where the variable {{EventName}} occurs in the message.



Step 6 - Review and Finish

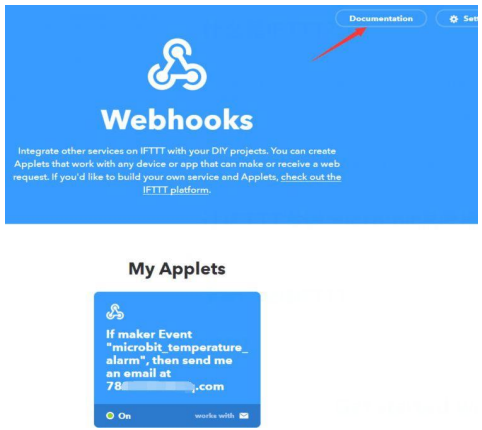
- Click Finish

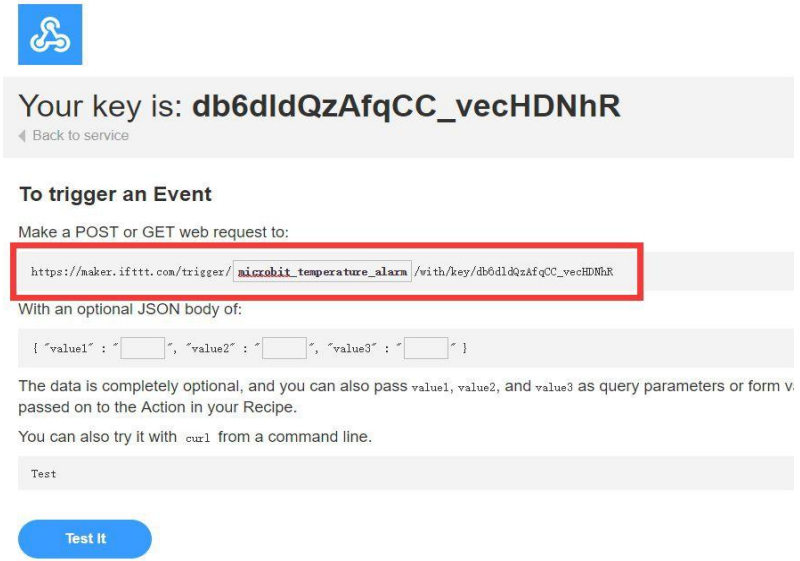




WebHook Information

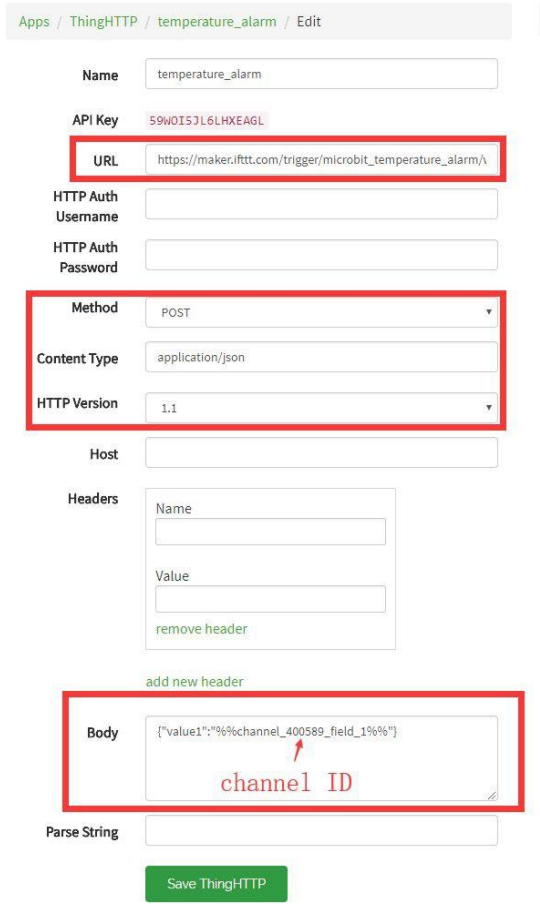
- Retrieve your Webhooks trigger information. Select My Applets > Services, and search for Webhooks.
- Select Webhooks, and then click the Documentation button





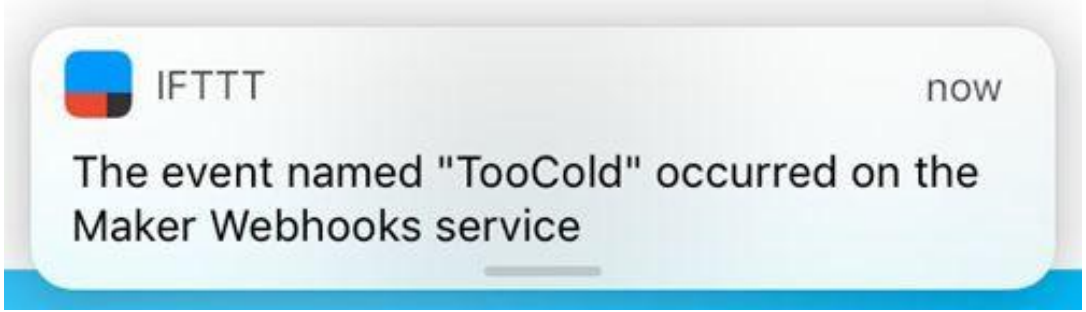
Create ThingHTTP

- To complete the trigger request, create a ThingHTTP.
- The ThingHTTP app lets you trigger predefined HTTP requests with an API key and a GET request from the web or from a device.
- Use ThingHTTP to trigger Webhooks at IFTTT.
- Choose Apps > ThingHTTP, and select New ThingHTTP.
- Edit your ThingHTTP settings.
  - Name - Name your ThingHTP.
  - URL - Enter the URL from the Webhooks documentation.
  - Method - Enter GET.
  - To send custom values in your message from IFTTT, add value1=YOUR_VALUE to the Body field. Change YOUR_VALUE to the value or text you want to send.
- Save the ThingHTTP.



Trigger Your Message

- Once the temperature in the channel reaches the set point for your React, you receive a notification in the IFTTT app on your device.

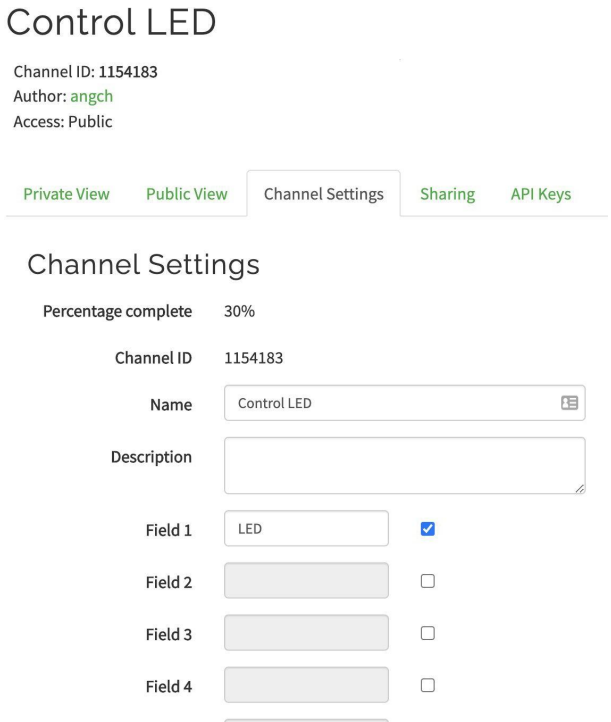


Activity: React to Environmental Data

- Setup a React on the Raspberry PI environment data
- Setup a React to trigger an email alert once the humidity data is above 50
- You need to setup ThingHTTP webhook and IFTTT for the email alert.

Control Device via IoT

- To control the Sense HAT LED from the cloud, setup a new channel - Control LED



Set LED Value

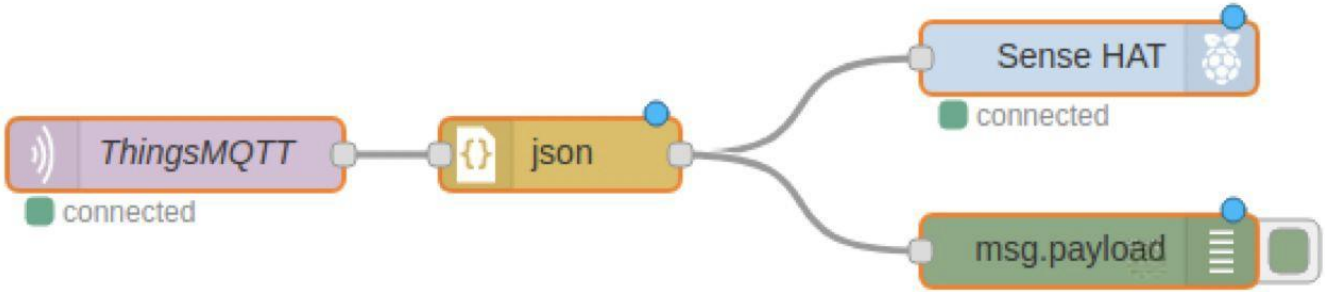
```
readChannelID =   1151501;
FieldID = 1;
readAPIKey = 'ZDHRC05UJUPEDEW0';
data = thingSpeakRead(readChannelID,'Fields',FieldID,'ReadKey',readAPIKey);
display(data)

LED = 0;
if (data > 30.00) LED =
    1;
else
    LED = 0;
end display(LED)

writeChannelID = 1154183; writeAPIKey =
'ME3X62ZYVC4LCDZY';
thingSpeakWrite(writeChannelID,LED,'WriteKey',writeAPIKey);
```

Setup Node-RED Flow

- Setup NODE-RED flow as follows to display the LED value on Sense HAT
- Json node is required to convert msg object string object to number.



Activity: Control Device from Cloud

- Setup a new channel to control Sense HAT LED
- Setup LED to 1 if the humidity is more than 40, else set the LED to 0

What is IoT Security?

- While traditional information cybersecurity revolves around software and how it is implemented, security for IoT adds an extra layer of complexity as the cyber and the physical worlds converge.
- A wide range of operational and maintenance scenarios in the IoT space rely on end-to-end device connectivity to enable users and services to interact, login, troubleshoot, send, or receive data from devices.
- Knowing what IoT security standards to adhere to is essential, because operational technology (OT) is too important and valuable to risk in the event of breaches, disasters, and other threats.

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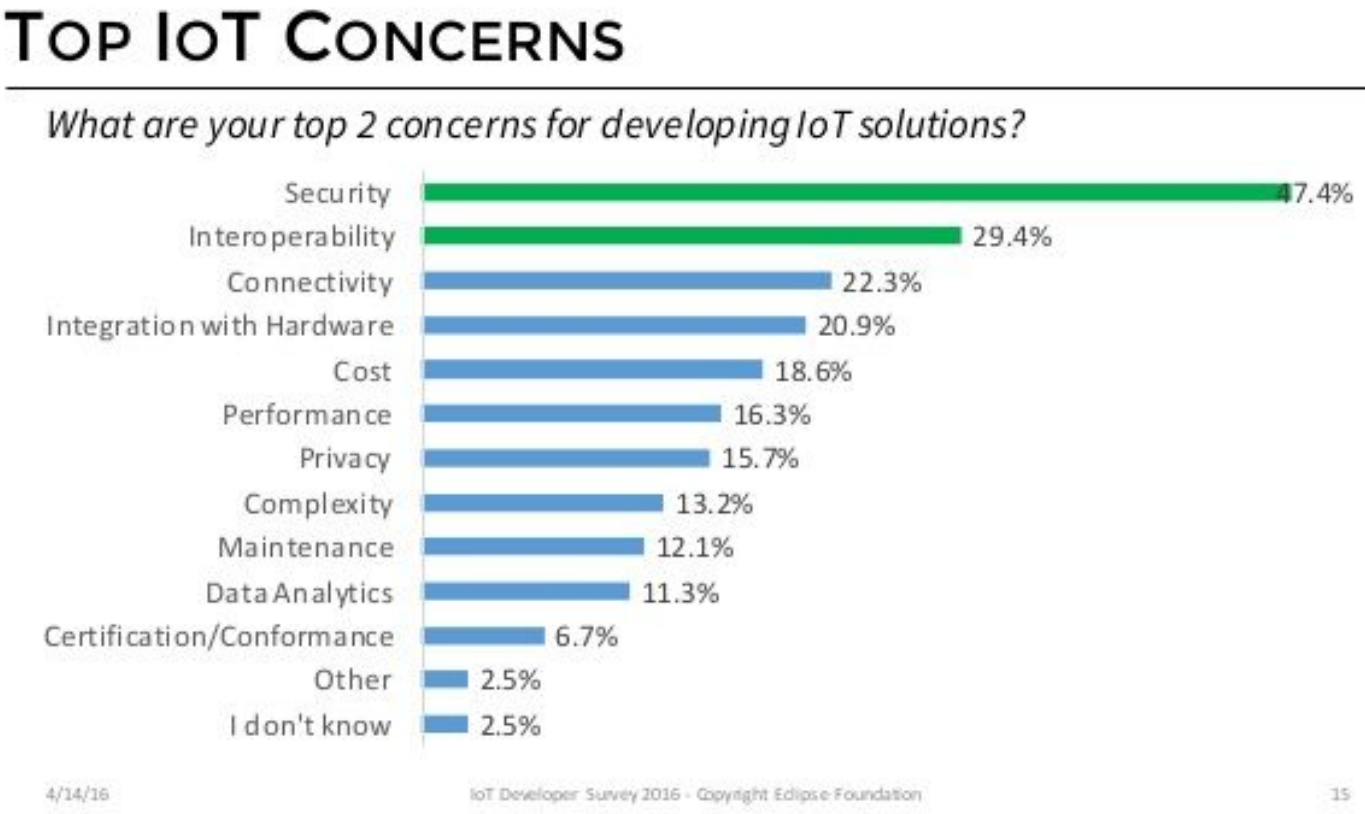
What's the Security Concern with IoT?



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What’s the Security Concern with IoT?

- Although IoT devices may seem too small or too specialized to be dangerous, there is real risk in what are really network-connected, general purpose computers that can be hijacked by attackers, resulting in problems beyond IoT security.
- Even the most mundane device can become dangerous when compromised over the internet—from spying with video baby monitors to interrupted services on life-saving health care equipment.
- Once attackers have control, they can steal data, disrupt delivery of services, or commit any other cybercrime they'd do with a computer.
- Attacks that compromise IoT infrastructure inflict damage, not just with data breaches and unreliable operations, but also physical harm to the facilities, or worse—to the humans operating or relying on those facilities.

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Threats of IoT Cybersecurity Attacks

- Processes → threats to processes both under your control, such as web services, and threats from external entities, such as users and satellite feeds, that interact with the system, but are not under the control of the application.
- Communication, also called data flows—threats around the communication path between devices, devices and field gateways, and device and cloud gateway.
- Storage —> threats to temporary data queues, operating systems (OS), and image storage

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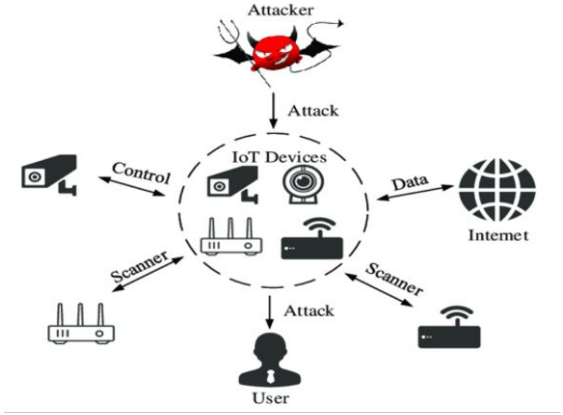
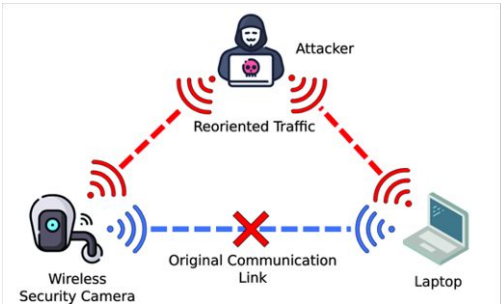
Spoofing

- An attacker can manipulate the state of a device anonymously.
- An attacker may intercept or partially override the broadcast and spoof the originator (often called man-in-the-middle or MitM attacks).
- An attacker can take advantages of the vulnerability of constrained or special-purpose devices. These devices, which often have one-for-all security facilities like password or PIN protection or rely on network shared key protections.
- When the shared secret to device or network (PIN, password, shared network key) is disclosed, it is possible to control the device or observe data emitted from the device.





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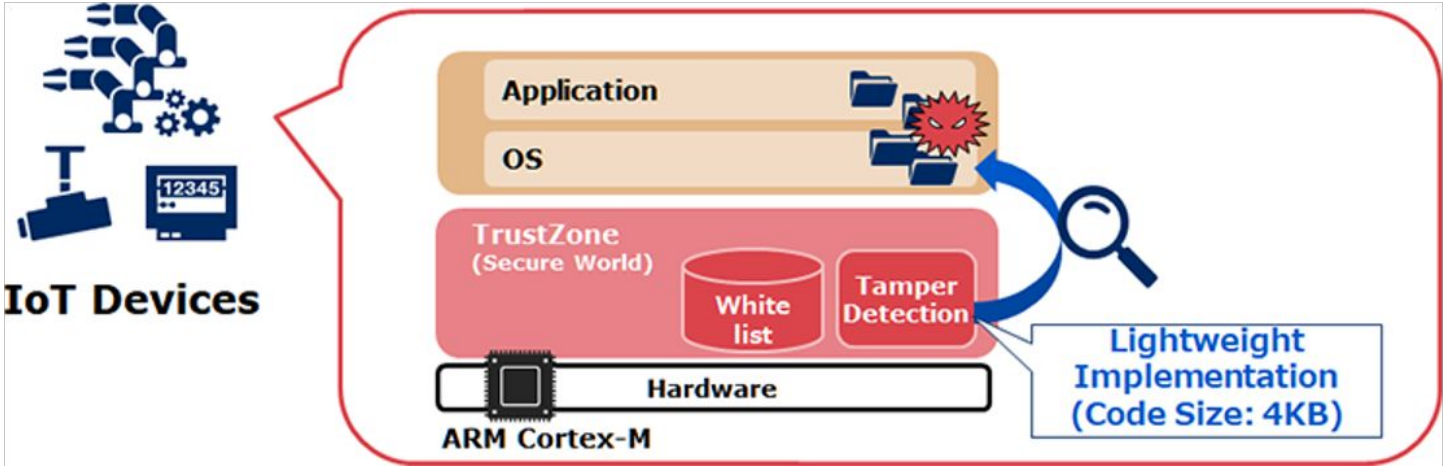
Tampering

- An attacker can tamper with any physical device → from battery drainage vulnerability or “sleep deprivation to random number generator (RNG) attacks made possible by freezing devices to reduce entropy.
- An attacker may partially or wholly replace the software running on the device, potentially allowing the replaced software to leverage the genuine identity of the device if the key material or the cryptographic facilities holding key materials were available to the illicit program.



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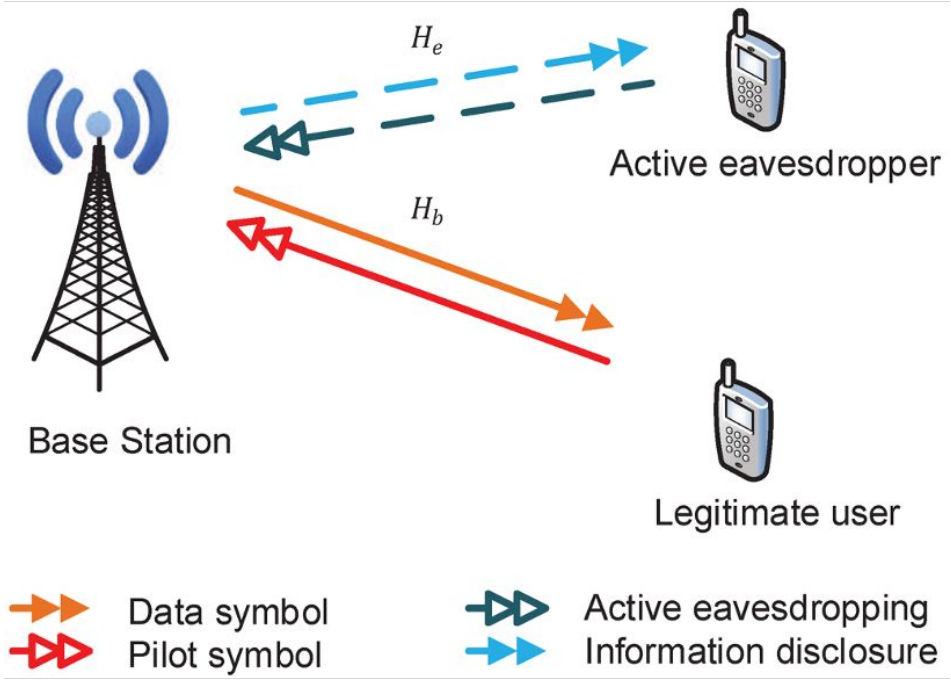
Eavesdrop

- An attacker may eavesdrop on a broadcast and obtain information without authorization or may jam the broadcast signal and deny information distribution.
- An attacker may intercept or partially override the broadcast and send false information.



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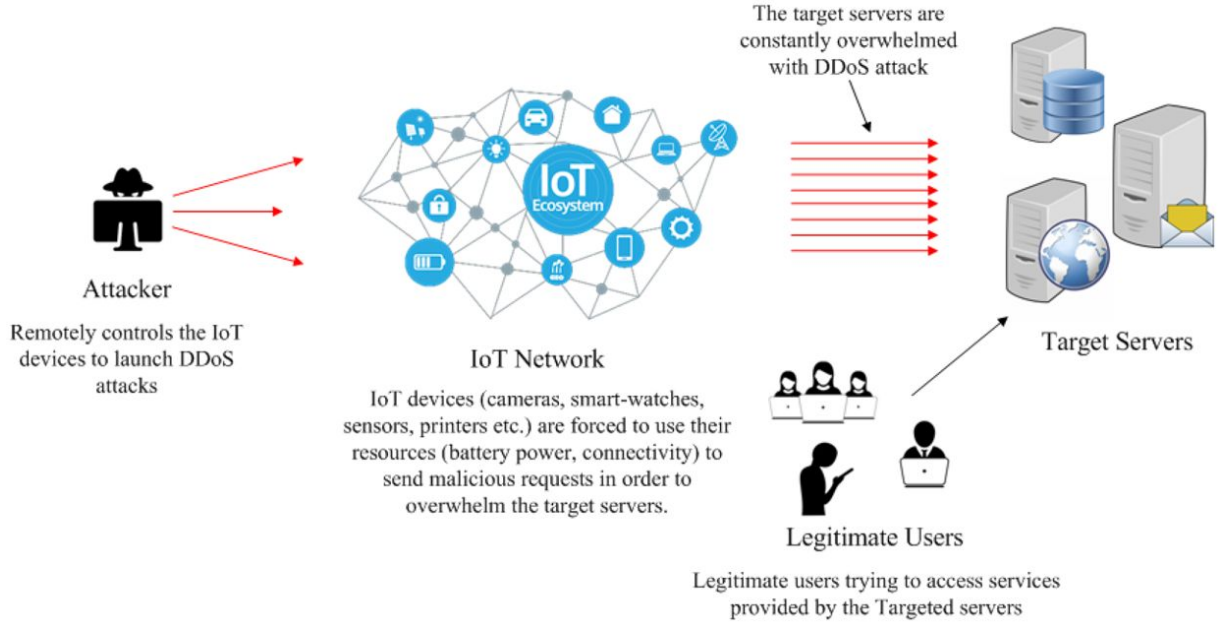
Denial of Service

- A device can be rendered incapable of functioning or communicating by interfering with radio frequencies or cutting wires.
- For example, a surveillance camera that had its power or network connection intentionally knocked out cannot report data, at all.



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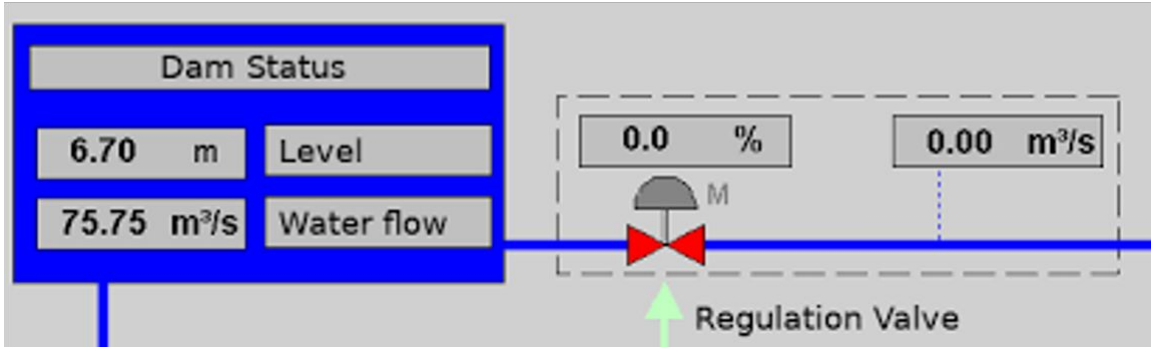
Elevation of Privilege

- A device that does specific function can be forced to do something else.
- For example, a valve that is programmed to open halfway can be tricked to open all the way.



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What Steps Can I Take to Secure IoT Deployments?

- Simplify security for IoT complexity
  - Integrate across teams and infrastructure to coordinate a comprehensive approach, from the physical devices and sensors to your data in the cloud.
- Prepare for IoT security specifically
  - Consider resource-constrained devices, geographic distribution of deployments, and the number of devices within an IoT security solution.
- Get smart about security analytics and remediation
  - Monitor everything connected to your IoT solution with security posture management. Stack rank the suggestions based on severity to decide what to fix first to reduce your risk. Make sure to have threat monitoring in place to get alerts and address IoT security threats quickly.
- Focus on customer and business data protection
  - By tracking all your connected data stores, admins, and other services that touch IoT, you can make sure your IoT apps are protected and your security for IoT is effective.

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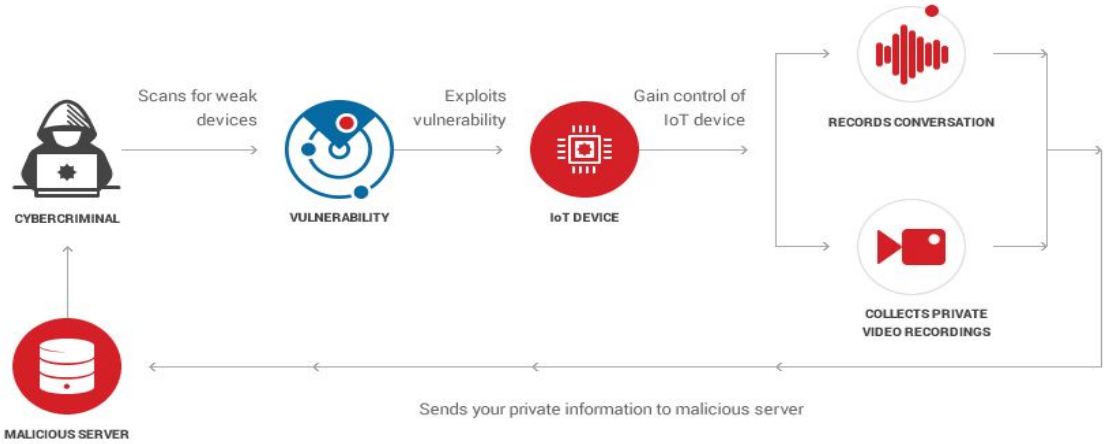
How to Secure IoT Devices in the Enterprise?

- Employ Device Discovery for Complete Visibility.
- Apply Network Segmentation for Stronger Defense.
- Adopt Secure Password Practices.
- Continue to Patch and Update Firmware When Available.
- Actively Monitor IoT Devices at All Times.



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Summary
Q&A

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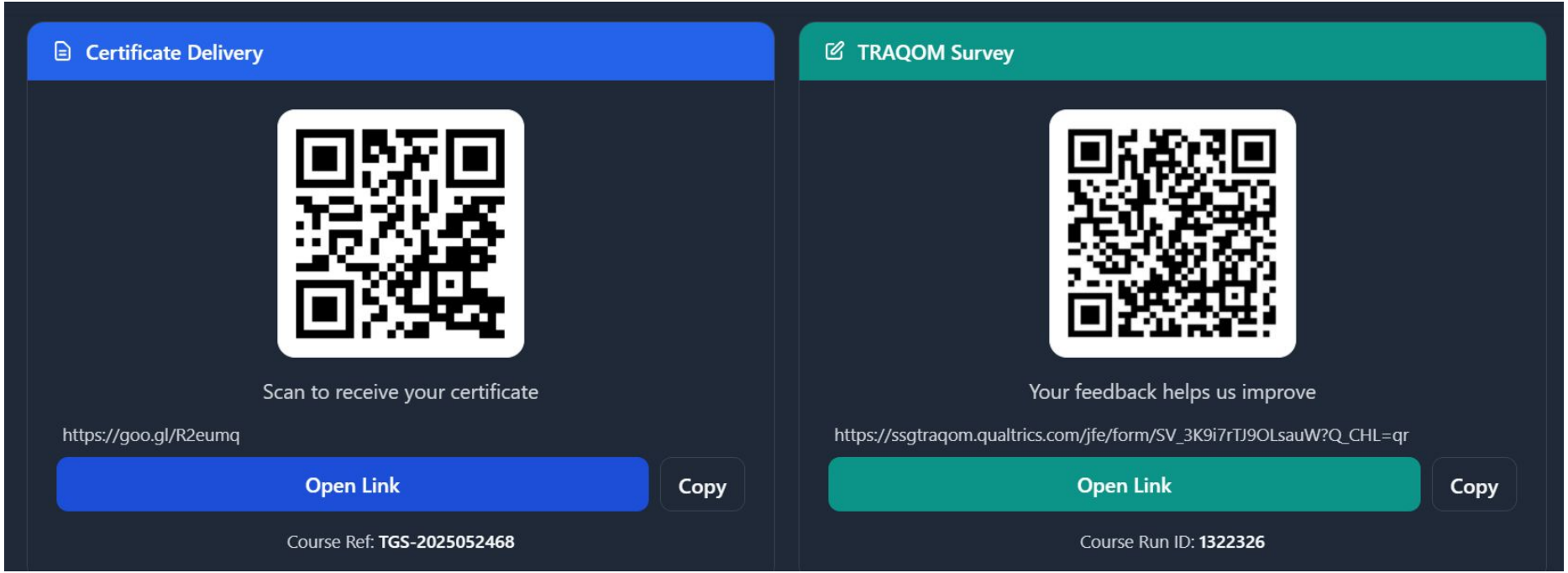

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Digital Attendance

- It is **mandatory** for you to take both AM, PM and Assessment digital attendance for WSQ funded courses.

- The trainer or administrator will show you the digital attendance QR code generated from SSG portal.

- Please scan the QR code from **your mobile phone camera** and submit your attendance.

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Final Assessment

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Recommended Courses

1. WSQ - Internet of Things (IoT) Fundamental for Beginners
2. WSQ - Computer Vision for Beginners
3. WSQ - Business Innovation with Internet-of-Things (IoT)
4. WSQ - Image and Video Processing with OpenCV
5. WSQ - Practical Electronics Design with Arduino Microcontroller

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Support

If you have any enquiries during and after the class, you can contact us below

Email: enquiry@tertiaryinfotech.com
Tel: 61000613

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